INCH-POUND
MIL-PRF-19978/1F
15 October 2009
SUPERSEDING
MIL-PRF-19978/1E
27 May 1999

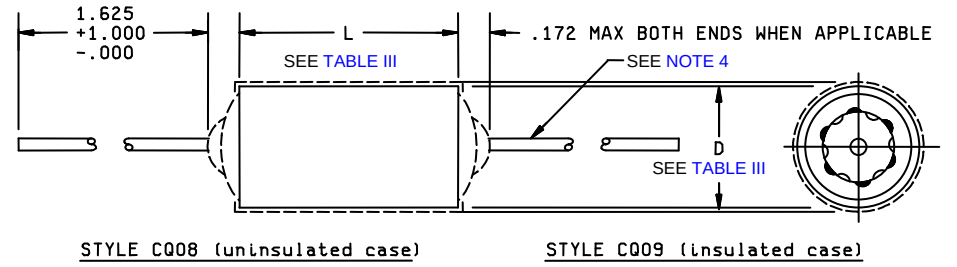
PERFORMANCE SPECIFICATION SHEET

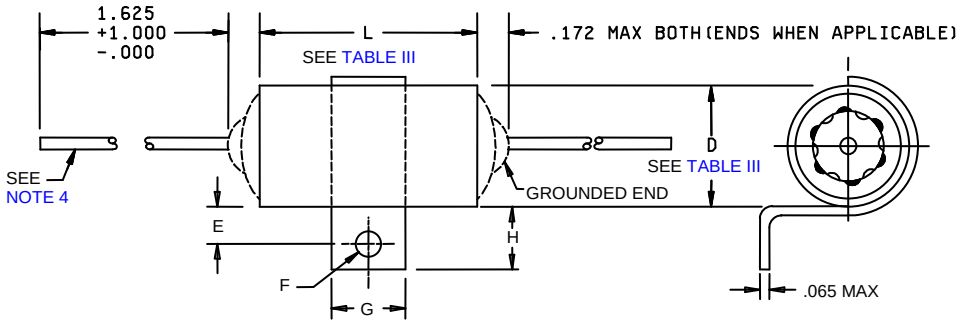
CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, (HERMETICALLY SEALED IN METAL CASES), STYLES CQ08, CQ09, CQ12, AND CQ13

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and MIL-PRF-19978.

CHARACTERISTICS K AND M ARE INACTIVE FOR NEW DESIGN AFTER 23 OCTOBER 1973. FOR REPLACEMENT, USE COMPARABLE CQR STYLES OF MIL-PRF-19978/9, MIL-PRF-19978/10, MIL-PRF-19978/13, MIL-PRF-19978/14, AND MIL-PRF-19978/15.



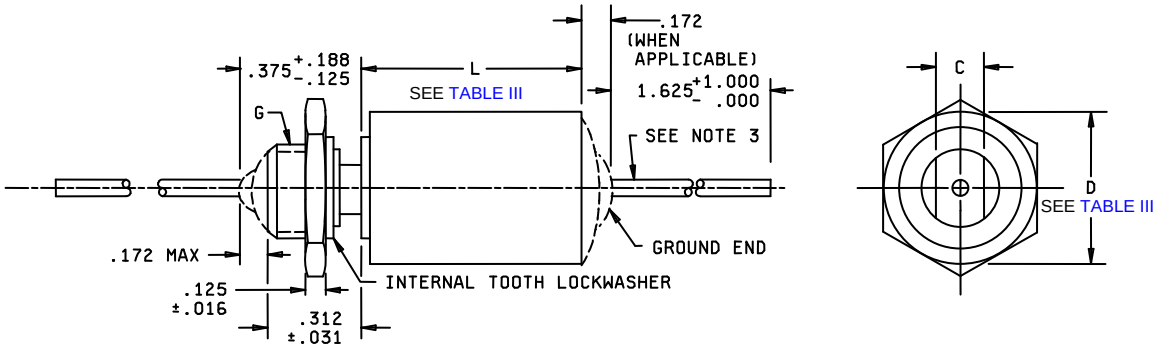


Case dimension D	H ±.062	G ±.062	E ±.031	F ±.005
Less than .500	.312	.250	.188	.144
.500 and Up	.438	.500	.250	.156

Style CQ12 (uninsulated case)

FIGURE 1. Styles CQ08, CQ09, CQ12, and CQ13 capacitors.

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Case dimension D	C ±.005	G
.562 or less	.250	5/16-24 UNF -2A
.670	.375	7/16-28 UNEF -2A
.750 and up	.437	1/2-28 UNEF -2A

Style CQ13 (uninsulated case)

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.144	3.66	.437	11.10
.004	0.10	.156	3.96	.438	11.13
.005	0.13	.172	4.37	.500	12.70
.016	0.41	.175	4.45	.562	14.27
.020	0.51	.188	4.78	.670	17.02
.025	0.64	.195	4.95	.750	19.05
.031	0.79	.235	5.97	1.000	25.40
.032	0.81	.250	6.35	1.562	39.67
.062	1.57	.312	7.92	1.625	41.28
.0625	1.59	.375	9.53		
.125	3.18	.400	10.16		

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Basic case dimensions for the insulated styles shall be the same as for the uninsulated styles.
4. Leads shall be of solder coated solid wire, .020 (No. 24 AWG) for cases .175 and .195 in diameter; .025 (No. 22 AWG) for cases .235 through .312 diameter; and .032 (No. 20 AWG) for cases .400 diameter and above except that style CQ13 capacitors shall be furnished with .032 (No. 20 AWG) lead wire. Tolerance on all lead wire diameters shall be +.004, -.001.
5. Capacitors with dimension L of 1.562 or D of .562 and larger, are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting such as tangential retainer for style CQ08, and a wrap-around hand for style CQ09. The supporting device will not be supplied with the capacitor. Style CQ13 shall be supplied with mounting nuts and lockwashers.
6. Lead length may be a minimum of 1-inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Styles CQ08, CQ09, CQ12 and CQ13 capacitors - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table III.

Case material: Nonmagnetic (screwneck and end seal may be of magnetic material).

Characteristic: See table I.

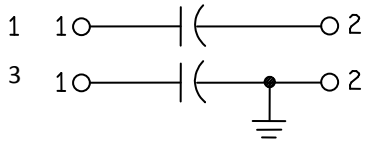
TABLE I. Characteristic.

Dielectric material	Characteristic symbol	Rated temperature	Dielectric absorption
Paper and polyethelene terephthalate	K 1/	-65° to +125°C	N/A
Polyethelene terephthalate	M 1/	-65° to +85°C	0.60
Polystyrene	P	-65° to +65°C	0.10
Polytetraflouroethylene	T	-65° to +170°C	0.10

1/ Characteristics K and M are inactive for new equipment design.

Terminals: Axial wire lead (see figure 1).

Circuit diagram:



Rated voltage: See table III.

Rated temperature: See table I.

Voltage derating (characteristic K only): See figure 2.

Capacitance (cap) (nom): See table III.

Capacitance tolerance: See table III.

Dissipation factor (DF):

Characteristic	Dissipation factor (max)
K	1.0 percent
M	0.6 percent
P and T	0.1 percent

Seal: Method 112 of MIL-STD-202, test condition letter A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-PRF-19978.

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Barometric pressure (reduced): In accordance with MIL-PRF-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Insulation resistance (IR):

Sleeving: In accordance with MIL-PRF-19978.

Terminal to terminal: See table II.

Terminal to case: In accordance with MIL-PRF-19978 except that characteristic K shall be greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt atmosphere (corrosion): In accordance with MIL-PRF-19978.

TABLE II. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/
Characteristic M	At 25°C
0 to 0.68 microfarad	100,000 megohms
Greater than 0.68 microfarad	68,000 megohm-microfarads 1/
	At 85°C
0 to 0.2 microfarad	25,000 megohms
Greater than 0.2 microfarad	5,000 megohm-microfarads 1/
Characteristic P	At 25°C
0 to 1.0 microfarad	100,000 megohms
Greater than 1.0 microfarad	100,000 megohm-microfarads 1/
	At 65°C
0 to 1.0 microfarad	40,000 megohms
Greater than 1.0 microfarad	40,000 megohm-microfarads 1/
Characteristic T	At 25°C
0 to 1.0 microfarad	100,000 megohms
Greater than 1.0 microfarad	100,000 megohm-microfarads 1/
	At 170°C
0 to1.0 microfarad	40,000 megohms
Greater than 1.0 microfarad	750,000 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

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Immersion:

DWV:

Sleeving: In accordance with MIL-PRF-19978.

IR:

Sleeving: In accordance with MIL-PRF-19978.

Solderability: In accordance with MIL-PRF-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

Moisture resistance:

DWV:

Sleeving: In accordance with MIL-PRF-19978.

IR:

Sleeving: In accordance with MIL-PRF-19978.

Dielectric absorption (not applicable to values of 0.01 μF or less): See table I.

Stability at low and high temperatures:

Low temperature:

Characteristic	Test temperature	Capacitance change (max)
K	-65°C +0°C, -5°C	-10 percent
M	-65°C +0°C, -5°C	-7 percent
T	-65°C +0°C, -5°C	+5 percent

High temperature:

Characteristic	Test temperature	Capacitance change (max)
K	+125°C +5°, -0°C	+10 percent
M	+85°C +5°, -0°C	+5 percent
T	+170°C +5°, -0°C	-5 percent

Temperature coefficient (characteristic P only): In accordance with MIL-PRF-19978.

Life:

Capacitance change:

Characteristic	Capacitance change (max) (of initial measured value)
K, M, and T	±5 percent
P (0.05 μF and less)	±6 percent
P (greater than 0.05 μF)	±3 percent

Resistance to soldering heat: In accordance with MIL-PRF-19978.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors.

Characteristic K				Case dimensions			
Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/))			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-KC392-3	200	.0039	G, J, K	.750	.235	.688	.235 2/
CQ--A-KC472-3	200	.0047	G, J, K	.750	.235	.688	.235 2/
CQ--A-KC562-3	200	.0056	G, J, K	.750	.235	.688	.235 2/
CQ--A-KC682-3	200	.0068	G, J, K	.750	.235	.688	.235 2/
CQ--A-KC183-3	200	.018	G, J, K	.875	.312	.812	.312 2/
CQ--A-KC223-3	200	.022	G, J, K	.875	.312	.812	.312 2/
CQ--A-KC273-3	200	.027	G, J, K	.875	.312	.812	.312 2/
CQ--A-KC333-3	200	.033	G, J, K	.875	.312	.812	.312 2/
CQ--A-KC393-3	200	.039	G, J, K	.875	.400	.812	.400
CQ--A-KC473-3	200	.047	G, J, K	.875	.400	.812	.400
CQ--A-KC563-3	200	.056	G, J, K	.875	.400	.812	.400
CQ--A-KC683-3	200	.068	G, J, K	.875	.400	.812	.400
CQ--A-KC823-3	200	.082	G, J, K	1.125	.400	1.062	.400
CQ--A-KC104-3	200	.10	G, J, K	1.125	.400	1.062	.400
CQ--A-KC124-3	200	.12	G, J, K	1.375	.400	1.312	.400
CQ--A-KC154-3	200	.15	G, J, K	1.375	.400	1.312	.400
CQ--A-KC184-3	200	.18	G, J, K	1.125	.562	1.062	.562
CQ--A-KC224-3	200	.22	G, J, K	1.125	.562	1.062	.562
CQ--A-KC274-3	200	.27	G, J, K	1.375	.562	1.312	.562
CQ--A-KC334-3	200	.33	G, J, K	1.375	.562	1.312	.562
CQ--A-KC394-3	200	.39	G, J, K	1.625	.562	1.562	.562
CQ--A-KC474-3	200	.47	G, J, K	1.625	.562	1.562	.562
CQ--A-KC564-3	200	.56	G, J, K	1.625	.670	1.562	.670
CQ--A-KC684-3	200	.68	G, J, K	1.625	.670	1.562	.670
CQ--A-KC824-3	200	.82	G, J, K	2.125	.750	2.062	.750
CQ--A-KC105-3	200	1.000	G, J, K	2.125	.750	2.062	.750
CQ--A-KE272-3	400	.0027	G, J, K	.750	.235	.688	.235 2/
CQ--A-KE332-3	400	.0033	G, J, K	.750	.235	.688	.235 2/
CQ--A-KE123-3	400	.012	G, J, K	.875	.312	.812	.312 2/
CQ--A-KE153-3	400	.015	G, J, K	.875	.312	.812	.312 2/
CQ--A-KE273-3	400	.027	G, J, K	.875	.400	.812	.400
CQ--A-KE333-3	400	.033	G, J, K	.875	.400	.812	.400
CQ--A-KE393-3	400	.039	G, J, K	1.125	.400	1.062	.400
CQ--A-KE473-3	400	.047	G, J, K	1.125	.400	1.062	.400
CQ--A-KE563-3	400	.056	G, J, K	1.375	.400	1.312	.400
CQ--A-KE683-3	400	.068	G, J, K	1.375	.400	1.312	.400
CQ--A-KE823-3	400	.082	G, J, K	1.125	.562	1.062	.562
CQ--A-KE104-3	400	.10	G, J, K	1.125	.562	1.062	.562
CQ--A-KE124-3	400	.12	G, J, K	1.375	.562	1.312	.562
CQ--A-KE154-3	400	.15	G, J, K	1.375	.562	1.312	.562
CQ--A-KE184-3	400	.18	G, J, K	1.625	.562	1.562	.562
CQ--A-KE224-3	400	.22	G, J, K	1.625	.562	1.562	.562
CQ--A-KE274-3	400	.27	G, J, K	1.625	.670	1.562	.670
CQ--A-KE334-3	400	.33	G, J, K	1.625	.670	1.562	.670
CQ--A-KE394-3	400	.39	G, J, K	2.125	.750	2.062	.750
CQ--A-KE474-3	400	.47	G, J, K	2.125	.750	2.062	.750

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Characteristic K				Case dimensions			
Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/))			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-KF102-3	600	.0010	G, J, K	.750	.235	.688	.235 2/
CQ--A-KF122-3	600	.0012	G, J, K	.750	.235	.688	.235 2/
CQ--A-KF152-3	600	.0015	G, J, K	.750	.235	.688	.235 2/
CQ--A-KF182-3	600	.0018	G, J, K	.750	.235	.688	.235 2/
CQ--A-KF222-3	600	.0022	G, J, K	.750	.235	.688	.235 2/
CQ--A-KF272-3	600	.0027	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF332-3	600	.0033	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF392-3	600	.0039	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF472-3	600	.0047	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF562-3	600	.0056	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF682-3	600	.0068	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF822-3	600	.0082	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF103-3	600	.010	G, J, K	.875	.312	.812	.312 2/
CQ--A-KF123-3	600	.012	G, J, K	.875	.400	.812	.400
CQ--A-KF153-3	600	.015	G, J, K	.875	.400	.812	.400
CQ--A-KF183-3	600	.018	G, J, K	.875	.400	.812	.400
CQ--A-KF223-3	600	.022	G, J, K	.875	.400	.812	.400
CQ--A-KF273-3	600	.027	G, J, K	1.125	.400	1.062	.400
CQ--A-KF333-3	600	.033	G, J, K	1.125	.400	1.062	.400
CQ--A-KF393-3	600	.039	G, J, K	1.375	.400	1.312	.400
CQ--A-KF473-3	600	.047	G, J, K	1.375	.400	1.312	.400
CQ--A-KF563-3	600	.056	G, J, K	1.125	.562	1.062	.562
CQ--A-KF683-3	600	.068	G, J, K	1.125	.562	1.062	.562
CQ--A-KF823-3	600	.082	G, J, K	1.375	.562	1.312	.562
CQ--A-KF104-3	600	.10	G, J, K	1.375	.562	1.312	.562
CQ--A-KF124-3	600	.12	G, J, K	1.625	.562	1.562	.562
CQ--A-KF154-3	600	.15	G, J, K	1.625	.562	1.562	.562
CQ--A-KF184-3	600	.18	G, J, K	1.625	.670	1.562	.670
CQ--A-KF224-3	600	.22	G, J, K	1.625	.670	1.562	.670
CQ--A-KF274-3	600	.27	G, J, K	2.125	.750	2.062	.750
CQ--A-KF334-3	600	.33	G, J, K	2.125	.750	2.062	.750
CQ--A-KF394-3	600	.39	G, J, K	2.375	.750	2.312	.750
CQ--A-KF474-3	600	.47	G, J, K	2.375	.750	2.312	.750
CQ--A-KG102-3 5/	1000	.0010	G, J, K	.875	.400	.812	.400
CQ--A-KG122-3 5/	1000	.0012	G, J, K	.875	.400	.812	.400
CQ--A-KG152-3 5/	1000	.0015	G, J, K	.875	.400	.812	.400
CQ--A-KG182-3 5/	1000	.0018	G, J, K	.875	.400	.812	.400
CQ--A-KG222-3 5/	1000	.0022	G, J, K	.875	.400	.812	.400
CQ--A-KG272-3 5/	1000	.0027	G, J, K	.875	.400	.812	.400
CQ--A-KG332-3 5/	1000	.0033	G, J, K	.875	.400	.812	.400
CQ--A-KG392-3 5/	1000	.0039	G, J, K	.875	.400	.812	.400
CQ--A-KG472-3 5/	1000	.0047	G, J, K	.875	.400	.812	.400
CQ--A-KG562-3 5/	1000	.0056	G, J, K	.875	.400	.812	.400
CQ--A-KG822-3 5/	1000	.0082	G, J, K	.875	.400	.812	.400
CQ--A-KG103-3 5/	1000	.010	G, J, K	.875	.400	.812	.400
CQ--A-KG682-3 5/	1000	.0068	G, J, K	.875	.400	.812	.400

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Case dimensions: Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/))			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
Characteristic K							
CQ--A-KG123-3 5/	1000	.012	G, J, K	1.125	.400	1.062	.400
CQ--A-KG153-3 5/	1000	.015	G, J, K	1.125	.400	1.062	.400
CQ--A-KG183-3 5/	1000	.018	G, J, K	1.375	.400	1.312	.400
CQ--A-KG223-3 5/	1000	.022	G, J, K	1.375	.400	1.312	.400
CQ--A-KG273-3 5/	1000	.027	G, J, K	1.125	.562	1.062	.562
CQ--A-KG333-3 5/	1000	.033	G, J, K	1.125	.562	1.062	.562
CQ--A-KG393-3 5/	1000	.039	G, J, K	1.375	.562	1.312	.562
CQ--A-KG473-3 5/	1000	.047	G, J, K	1.375	.562	1.312	.562
CQ--A-KG563-3 5/	1000	.056	G, J, K	1.625	.562	1.562	.562
CQ--A-KG683-3 5/	1000	.068	G, J, K	1.625	.562	1.562	.562
CQ--A-KG823-3 5/	1000	.082	G, J, K	1.625	.670	1.562	.670
CQ--A-KG104-3 5/	1000	.10	G, J, K	1.625	.670	1.562	.670
CQ--A-KG124-3 5/	1000	.12	G, J, K	1.875	.670	1.812	.670
CQ--A-KG154-3 5/	1000	.15	G, J, K	1.875	.670	1.812	.670
CQ--A-KG184-3	1000	.18	G, J, K	2.125	.750	2.062	.750
CQ--A-KG224-3	1000	.22	G, J, K	2.125	.750	2.062	.750
CQ--A-KG274-3	1000	.27	G, J, K	2.125	1.0	2.062	1.000
CQ--A-KG334-3	1000	.33	G, J, K	2.125	1.0	2.062	1.000
CQ--A-KG394-3	1000	.39	G, J, K	2.375	1.0	2.312	1.000
CQ--A-KG474-3	1000	.47	G, J, K	2.375	1.0	2.312	1.000
Characteristic M				Case dimensions			
CQ--A-MZ155-3	30	1.5	G, J, K	1.625	.562	1.531	.562
CQ--A-MZ225-3	30	2.2	G, J, K	1.875	.670	1.781	.670
CQ--A-MZ335-3	30	3.3	G, J, K	1.875	.750	1.781	.750
CQ--A-MZ475-3	30	4.7	G, J, K	2.625	.750	2.531	.750
CQ--A-MZ685-3	30	6.8	G, J, K	2.125	1.000	2.031	1.000
CQ--A-MZ106-3	30	10.0	G, J, K	2.375	1.000	2.281	1.000
CQ--A-MA102-3	50	.0010	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA122-3	50	.0012	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA152-3	50	.0015	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA182-3	50	.0018	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA222-3	50	.0022	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA272-3	50	.0027	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA332-3	50	.0033	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA392-3	50	.0039	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA472-3	50	.0047	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA562-3	50	.0056	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA682-3	50	.0068	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA822-3	50	.0082	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA103-3	50	.010	G, J, K	.500	.175	.406	.175 2/
CQ--A-MA123-3	50	.012	G, J, K	.500	.235	.406	.235 2/
CQ--A-MA153-3	50	.015	G, J, K	.500	.235	.406	.235 2/
CQ--A-MA183-3	50	.018	G, J, K	.500	.235	.406	.235 2/
CQ--A-MA223-3	50	.022	G, J, K	.500	.235	.406	.235 2/
CQ--A-MA273-3	50	.027	G, J, K	.500	.312	.406	.312 2/
CQ--A-MA333-3	50	.033	G, J, K	.500	.312	.406	.312 2/

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Characteristic M				Case dimensions			
Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/))			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-MA393-3	50	.039	G, J, K	.500	.312	.406	.312 2/
CQ--A-MA473-3	50	.047	G, J, K	.500	.312	.406	.312 2/
CQ--A-MA563-3	50	.056	G, J, K	.562	.312	.468	.312 2/
CQ--A-MA683-3	50	.068	G, J, K	.562	.312	.468	.312 2/
CQ--A-MA124-3	50	.12	G, J, K	.875	.312	.781	.312 2/
CQ--A-MA154-3	50	.15	G, J, K	.875	.312	.781	.312 2/
CQ--A-MA184-3	50	.18	G, J, K	.875	.400	.781	.400
CQ--A-MA224-3	50	.22	G, J, K	.875	.400	.781	.400
CQ--A-MA274-3	50	.27	G, J, K	1.000	.400	.906	.400
CQ--A-MA334-3	50	.33	G, J, K	1.000	.400	.906	.400
CQ--A-MA394-3	50	.39	G, J, K	1.125	.400	1.031	.400
CQ--A-MA474-3	50	.47	G, J, K	1.125	.400	1.031	.400
CQ--A-MA564-3	50	.56	G, J, K	1.375	.400	1.281	.400
CQ--A-MA684-3	50	.68	G, J, K	1.375	.400	1.281	.400
CQ--A-MA824-3	50	.82	G, J, K	1.125	.562	1.031	.562
CQ--A-MA105-3	50	1.0	G, J, K	1.125	.562	1.031	.562
CQ--A-MA125-3	50	1.2	G, J, K	1.375	.562	1.281	.562
CQ--A-MB125-3	100	1.2	G, J, K	1.625	.670	1.531	.670
CQ--A-MB155-3	100	1.5	G, J, K	1.625	.670	1.531	.670
CQ--A-MB185-3	100	1.8	G, J, K	1.625	.670	1.531	.670
CQ--A-MB225-3	100	2.2	G, J, K	1.875	.750	1.781	.750
CQ--A-MB275-3	100	2.7	G, J, K	2.125	.750	2.031	.750
CQ--A-MB335-3	100	3.3	G, J, K	2.375	.750	2.281	.750
CQ--A-MB396-3	100	3.9	G, J, K	1.875	1.000	1.781	1.000
CQ--A-MB475-3	100	4.7	G, J, K	2.125	1.000	2.031	1.000
CQ--A-MB565-3	100	5.6	G, J, K	2.375	1.000	2.281	1.000
CQ--A-MB685-3	100	6.8	G, J, K	2.625	1.000	2.531	1.000
CQ--A-MC102-3	200	.0010	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC122-3	200	.0012	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC152-3	200	.0015	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC182-3	200	.0018	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC222-3	200	.0022	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC272-3	200	.0027	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC332-3	200	.0033	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC392-3	200	.0039	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC472-3	200	.0047	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC562-3	200	.0056	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC682-3	200	.0068	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC822-3	200	.0082	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC103-3	200	.010	G, J, K	.750	.175	.656	.175 2/
CQ--A-MC123-3	200	.012	G, J, K	.719	.195	.625	.195 2/
CQ--A-MC153-3	200	.015	G, J, K	.719	.195	.625	.195 2/
CQ--A-MC183-3	200	.018	G, J, K	.719	.235	.625	.235 2/
CQ--A-MC223-3	200	.022	G, J, K	.719	.235	.625	.235 2/
CQ--A-MC273-3	200	.027	G, J, K	.719	.235	.625	.235 2/
CQ--A-MC333-3	200	.033	G, J, K	.719	.235	.625	.235 2/

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Characteristic M				Case dimensions			
Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/))			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-MC393-3	200	.039	G, J, K	.719	.312	.625	.312 2/
CQ--A-MC473-3	200	.047	G, J, K	.719	.312	.625	.312 2/
CQ--A-MC563-3	200	.056	G, J, K	.844	.312	.750	.312 2/
CQ--A-MC683-3	200	.068	G, J, K	.844	.312	.750	.312 2/
CQ--A-MC823-3	200	.082	G, J, K	.844	.312	.750	.312 2/
CQ--A-MC104-3	200	.10	G, J, K	.844	.312	.750	.312 2/
CQ--A-MC124-3	200	.12	G, J, K	.812	.400	.718	.400
CQ--A-MC154-3	200	.15	G, J, K	.812	.400	.718	.400
CQ--A-MC184-3	200	.18	G, J, K	1.062	.400	.969	.400
CQ--A-MC224-3	200	.22	G, J, K	1.062	.400	.969	.400
CQ--A-MC274-3	200	.27	G, J, K	1.250	.400	1.156	.400
CQ--A-MC334-3	200	.33	G, J, K	1.250	.400	1.156	.400
CQ--A-MC394-3	200	.39	G, J, K	1.188	.500	1.094	.500
CQ--A-MC474-3	200	.47	G, J, K	1.188	.500	1.094	.500
CQ--A-MC564-3	200	.56	G, J, K	1.375	.562	1.281	.562
CQ--A-MC684-3	200	.68	G, J, K	1.375	.562	1.281	.562
CQ--A-MC824-3	200	.82	G, J, K	1.875	.562	1.781	.562
CQ--A-MC105-3	200	1.0	G, J, K	1.875	.562	1.781	.562
CQ--A-ME102-3	400	.0010	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME122-3	400	.0012	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME152-3	400	.0015	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME182-3	400	.0018	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME222-3	400	.0022	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME272-3	400	.0027	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME332-3	400	.0033	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME392-3	400	.0039	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME472-3	400	.0047	G, J, K	.812	.235	.718	.235 2/
CQ--A-ME562-3	400	.0056	G, J, K	.938	.312	.844	.312 2/
CQ--A-ME682-3	400	.0068	G, J, K	.938	.312	.844	.312 2/
CQ--A-ME822-3	400	.0082	G, J, K	.938	.312	.844	.312 2/
CQ--A-ME103-3	400	.010	G, J, K	.938	.312	.844	.312 2/
CQ--A-ME123-3	400	.012	G, J, K	.938	.312	.844	.312 2/
CQ--A-ME153-3	400	.015	G, J, K	.938	.312	.844	.312 2/
CQ--A-ME183-3	400	.018	G, J, K	1.062	.312	.969	.312 2/
CQ--A-ME223-3	400	.022	G, J, K	1.062	.312	.969	.312 2/
CQ--A-ME273-3	400	.027	G, J, K	.938	.400	.844	.400
CQ--A-ME333-3	400	.033	G, J, K	.938	.400	.844	.400
CQ--A-ME393-3	400	.039	G, J, K	1.062	.400	.969	.400
CQ--A-ME473-3	400	.047	G, J, K	1.062	.400	.969	.400
CQ--A-ME563-3	400	.056	G, J, K	1.250	.400	1.156	.400
CQ--A-ME683-3	400	.068	G, J, K	1.250	.400	1.156	.400
CQ--A-ME823-3	400	.082	G, J, K	1.562	.400	1.469	.400
CQ--A-ME104-3	400	.10	G, J, K	1.562	.400	1.469	.400
CQ--A-ME124-3	400	.12	G, J, K	1.281	.562	1.188	.562
CQ--A-ME154-3	400	.15	G, J, K	1.281	.562	1.188	.562
CQ--A-ME184-3	400	.18	G, J, K	1.531	.562	1.438	.562

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Characteristic M				Case dimensions			
Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/))			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-ME224-3	400	.22	G, J, K	1.531	.562	1.438	.562
CQ--A-ME274-3	400	.27	G, J, K	1.906	.562	1.812	.562
CQ--A-ME334-3	400	.33	G, J, K	1.906	.562	1.812	.562
CQ--A-ME394-3	400	.39	G, J, K	1.906	.670	1.812	.670
CQ--A-ME474-3	400	.47	G, J, K	1.906	.670	1.812	.670
CQ--A-ME564-3	400	.56	G, J, K	2.156	.750	2.062	.750
CQ--A-ME684-3	400	.68	G, J, K	2.156	.750	2.062	.750
CQ--A-ME824-3	400	.82	G, J, K	2.656	.750	2.562	.750
CQ--A-ME105-3	400	1.0	G, J, K	2.656	.750	2.562	.750
CQ--A-MF102-3	600	.0010	G, J, K	.812	.235	.718	.235 2/
CQ--A-MF122-3	600	.0012	G, J, K	.812	.235	.718	.235 2/
CQ--A-MF152-3	600	.0015	G, J, K	.812	.235	.718	.235 2/
CQ--A-MF182-3	600	.0018	G, J, K	.812	.235	.718	.235 2/
CQ--A-MF222-3	600	.0022	G, J, K	.812	.235	.718	.235 2/
CQ--A-MF272-3	600	.0027	G, J, K	.812	.235	.718	.235 2/
CQ--A-MF332-3	600	.0033	G, J, K	.938	.312	.844	.312 2/
CQ--A-MF392-3	600	.0039	G, J, K	.938	.312	.844	.312 2/
CQ--A-MF472-3	600	.0047	G, J, K	.938	.312	.844	.312 2/
CQ--A-MF562-3	600	.0056	G, J, K	.938	.312	.844	.312 2/
CQ--A-MF682-3	600	.0068	G, J, K	.938	.312	.844	.312 2/
CQ--A-MF822-3	600	.0082	G, J, K	1.062	.312	.969	.312 2/
CQ--A-MF103-3	600	.010	G, J, K	1.062	.312	.969	.312 2/
CQ--A-MF123-3	600	.012	G, J, K	.938	.400	.844	.400
CQ--A-MF153-3	600	.015	G, J, K	.938	.400	.844	.400
CQ--A-MF183-3	600	.018	G, J, K	1.062	.400	.969	.400
CQ--A-MF223-3	600	.022	G, J, K	1.062	.400	.969	.400
CQ--A-MF273-3	600	.027	G, J, K	1.312	.400	1.219	.400
CQ--A-MF333-3	600	.033	G, J, K	1.312	.400	1.219	.400
CQ--A-MF393-3	600	.039	G, J, K	1.562	.400	1.469	.400
CQ--A-MF473-3	600	.047	G, J, K	1.562	.400	1.469	.400
CQ--A-MF563-3	600	.056	G, J, K	1.250	.562	1.156	.562
CQ--A-MF683-3	600	.068	G, J, K	1.250	.562	1.156	.562
CQ--A-MF823-3	600	.082	G, J, K	1.500	.562	1.406	.562
CQ--A-MF104-3	600	.10	G, J, K	1.500	.562	1.406	.562
CQ--A-MF124-3	600	.12	G, J, K	1.875	.562	1.781	.562
CQ--A-MF154-3	600	.15	G, J, K	1.875	.562	1.781	.562
CQ--A-MF184-3	600	.18	G, J, K	1.875	.670	1.781	.670
CQ--A-MF224-3	600	.22	G, J, K	1.875	.670	1.781	.670
CQ--A-MF274-3	600	.27	G, J, K	2.125	.750	2.031	.750
CQ--A-MF334-3	600	.33	G, J, K	2.125	.750	2.031	.750
CQ--A-MF394-3	600	.39	G, J, K	2.625	.750	2.531	.750
CQ--A-MF474-3	600	.47	G, J, K	2.625	.750	2.531	.750
CQ--A-MF564-3	600	.56	G, J, K	2.125	1.000	2.031	1.000
CQ--A-MF684-3	600	.68	G, J, K	2.125	1.000	2.031	1.000
CQ--A-MG102-3 5/	1000	.0010	G, J, K	.938	.312	.844	.312 2/
CQ--A-MG122-3 5/	1000	.0012	G, J, K	.938	.312	.844	.312 2/

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Case dimensions: Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/))			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-MG152-3 5/	1000	.0015	G, J, K	.938	.312	.844	.312 2/
CQ--A-MG182-3 5/	1000	.0018	G, J, K	.938	.312	.844	.312 2/
CQ--A-MG222-3 5/	1000	.0022	G, J, K	.938	.312	.844	.312 2/
CQ--A-MG272-3 5/	1000	.0027	G, J, K	.938	.312	.844	.312 2/
CQ--A-MG332-3 5/	1000	.0033	G, J, K	.938	.312	.844	.312 2/
CQ--A-MG392-3 5/	1000	.0039	G, J, K	.938	.400	.844	.400
CQ--A-MG472-3 5/	1000	.0047	G, J, K	.938	.400	.844	.400
CQ--A-MG562-3 5/	1000	.0056	G, J, K	1.062	.400	.969	.400
CQ--A-MG682-3 5/	1000	.0068	G, J, K	1.062	.400	.969	.400
CQ--A-MG822-3 5/	1000	.0082	G, J, K	1.312	.400	1.219	.400
CQ--A-MG103-3 5/	1000	.010	G, J, K	1.312	.400	1.219	.400
CQ--A-MG123-3 5/	1000	.012	G, J, K	1.562	.400	1.469	.400
CQ--A-MG183-3 5/	1000	.018	G, J, K	1.562	.400	1.469	.400
CQ--A-MG223-3 5/	1000	.022	G, J, K	1.250	.562	1.156	.562
CQ--A-MG273-3 5/	1000	.027	G, J, K	1.250	.562	1.156	.562
CQ--A-MG333-3 5/	1000	.033	G, J, K	1.500	.562	1.406	.562
CQ--A-MG393-3 5/	1000	.039	G, J, K	1.875	.562	1.781	.562
CQ--A-MG473-3 5/	1000	.047	G, J, K	1.875	.562	1.781	.562
CQ--A-MG563-3 5/	1000	.056	G, J, K	1.875	.670	1.781	.670
CQ--A-MG683-3 5/	1000	.068	G, J, K	1.875	.670	1.781	.670
CQ--A-MG823-3	1000	.082	G, J, K	2.125	.750	2.031	.750
CQ--A-MG104-3	1000	.10	G, J, K	2.125	.750	2.031	.750
CQ--A-MG124-3	1000	.12	G, J, K	1.875	1.000	1.781	1.000
CQ--A-MG154-3	1000	.15	G, J, K	1.875	1.000	1.781	1.000
CQ--A-MG184-3	1000	.18	G, J, K	2.375	1.000	2.281	1.000
CQ--A-MG224-3	1000	.22	G, J, K	2.375	1.000	2.281	1.000
Characteristic P				Case dimensions			
CQ--A-PA102-3	50	.0010	F, J, K	.749	.235	.687	.235 2/
CQ--A-PA122-3	50	.0012	F, J, K	.749	.235	.687	.235 2/
CQ--A-PA152-3	50	.0015	F, J, K	.749	.235	.687	.235 2/
CQ--A-PA182-3	50	.0018	F, J, K	.749	.235	.687	.235 2/
CQ--A-PA222-3	50	.0022	F, J, K	.749	.235	.687	.235 2/
CQ--A-PA272-3	50	.0027	F, J, K	.875	.235	.812	.235 2/
CQ--A-PA332-3	50	.0033	F, J, K	.875	.235	.812	.235 2/
CQ--A-PA392-3	50	.0039	F, J, K	.875	.235	.812	.235 2/
CQ--A-PA472-3	50	.0047	F, J, K	.875	.235	.812	.235 2/
CQ--A-PA562-3	50	.0056	F, J, K	.875	.312	.812	.312 2/
CQ--A-PA682-3	50	.0068	F, J, K	.875	.312	.812	.312 2/
CQ--A-PA822-3	50	.0082	F, J, K	.875	.312	.812	.312 2/
CQ--A-PA103-3	50	.010	F, J, K	.875	.312	.812	.312 2/
CQ--A-PA123-3	50	.012	F, J, K	.875	.312	.812	.312 2/
CQ--A-PA153-3	50	.015	F, J, K	.875	.312	.812	.312 2/
CQ--A-PA183-3	50	.018	F, J, K	.875	.400	.812	.400
CQ--A-PA223-3	50	.022	F, J, K	.875	.400	.812	.400
CQ--A-PA273-3	50	.027	F, J, K	1.125	.400	1.062	.400
CQ--A-PA333-3	50	.033	F, J, K	1.125	.400	1.062	.400

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Characteristic P				Case dimensions			
Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/))			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-PA393-3	50	.039	F, J, K	1.125	.400	1.062	.400
CQ--A-PA473-3	50	.047	F, J, K	1.125	.400	1.062	.400
CQ--A-PA563-3	50	.056	F, J, K	1.375	.400	1.312	.400
CQ--A-PA683-3	50	.068	F, J, K	1.375	.400	1.312	.400
CQ--A-PA823-3	50	.082	F, J, K	1.125	.562	1.062	.400
CQ--A-PA104-3	50	.10	F, J, K	1.125	.562	1.062	.400
CQ--A-PA124-3	50	.12	F, J, K	1.375	.562	1.312	.562
CQ--A-PA154-3	50	.15	F, J, K	1.375	.562	1.312	.562
CQ--A-PA184-3	50	.18	F, J, K	1.625	.562	1.562	.562
CQ--A-PA224-3	50	.22	F, J, K	1.625	.562	1.562	.562
CQ--A-PA274-3	50	.27	F, J, K	1.625	.670	1.562	.670
CQ--A-PA334-3	50	.33	F, J, K	1.625	.670	1.562	.670
CQ--A-PA394-3	50	.39	F, J, K	2.125	.750	2.062	.750
CQ--A-PA474-3	50	.47	F, J, K	2.125	.750	2.062	.750
CQ--A-PA564-3	50	.56	F, J, K	2.125	.750	2.062	.750
CQ--A-PA684-3	50	.68	F, J, K	2.125	.750	2.062	.750
CQ--A-PA824-3	50	.82	F, J, K	2.125	1.000	2.062	1.000
CQ--A-PA105-3	50	1.0	F, J, K	2.125	1.000	2.062	1.000
CQ--A-PC222-3	200	.0022	F, J, K	.875	.235	.812	.235 2/
CQ--A-PC332-3	200	.0033	F, J, K	.875	.312	.812	.312 2/
CQ--A-PC472-3	200	.0047	F, J, K	.875	.312	.812	.312 2/
CQ--A-PC682-3	200	.0068	F, J, K	.875	.400	.812	.400
CQ--A-PC103-3	200	.010	F, J, K	1.125	.400	1.062	.400
CQ--A-PC153-3	200	.015	F, J, K	1.125	.400	1.062	.400
CQ--A-PC223-3	200	.022	F, J, K	1.375	.400	1.312	.400
CQ--A-PC333-3	200	.033	F, J, K	1.125	.562	1.062	.562
CQ--A-PC473-3	200	.047	F, J, K	1.375	.562	1.312	.562
CQ--A-PC683-3	200	.068	F, J, K	1.625	.562	1.562	.562
CQ--A-PC104-3	200	.10	F, J, K	1.625	.670	1.562	.670
CQ--A-PC154-3	200	.15	F, J, K	2.125	.750	2.062	.750
CQ--A-PC224-3	200	.22	F, J, K	2.125	.750	2.062	.750
CQ--A-PC334-3	200	.33	F, J, K	2.125	1.000	2.062	1.000
CQ--A-PC474-3	200	.47	F, J, K	2.125	1.000	2.062	1.000
CQ--A-PE102-3	400	.0010	F, J, K	.875	.235	.812	.235 2/
CQ--A-PE152-3	400	.0015	F, J, K	.875	.312	.812	.312 2/
CQ--A-PE222-3	400	.0022	F, J, K	.875	.312	.812	.312 2/
CQ--A-PE332-3	400	.0033	F, J, K	.875	.312	.812	.312 2/
CQ--A-PE472-3	400	.0047	F, J, K	.875	.400	.812	.400
CQ--A-PE682-3	400	.0068	F, J, K	1.125	.400	1.062	.400
CQ--A-PE103-3	400	.010	F, J, K	1.125	.400	1.062	.400
CQ--A-PE153-3	400	.015	F, J, K	1.375	.400	1.312	.400
CQ--A-PE223-3	400	.022	F, J, K	1.125	.562	1.062	.562
CQ--A-PE333-3	400	.033	F, J, K	1.375	.562	1.312	.562
CQ--A-PE473-3	400	.047	F, J, K	1.625	.562	1.562	.562
CQ--A-PE683-3	400	.068	F, J, K	1.625	.670	1.562	.670
CQ--A-PE104-3	400	.10	F, J, K	2.125	.750	2.062	.750

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Characteristic P				Case dimensions			
Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/)			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-PE154-3	400	.15	F, J, K	2.125	.750	2.062	.750
CQ--A-PF102-3	600	.0010	F, J, K	.875	.400	.812	.400
CQ--A-PF152-3	600	.0015	F, J, K	1.125	.400	1.062	.400
CQ--A-PF222-3	600	.0022	F, J, K	1.125	.400	1.062	.400
CQ--A-PF332-3	600	.0033	F, J, K	1.125	.400	1.062	.400
CQ--A-PF472-3	600	.0047	F, J, K	1.125	.400	1.062	.400
CQ--A-PF682-3	600	.0068	F, J, K	1.125	.400	1.062	.400
CQ--A-PF103-3	600	.010	F, J, K	1.375	.400	1.312	.400
CQ--A-PF153-3	600	.015	F, J, K	1.375	.400	1.312	.400
CQ--A-PF223-3	600	.022	F, J, K	1.375	.562	1.312	.562
CQ--A-PF333-3	600	.033	F, J, K	1.375	.562	1.312	.562
CQ--A-PF473-3	600	.047	F, J, K	1.625	.670	1.562	.670
CQ--A-PF683-3	600	.068	F, J, K	2.125	.750	2.062	.750
CQ--A-PF104-3	600	.10	F, J, K	2.125	.750	2.062	.750
CQ--A-PF154-3	600	.15	F, J, K	2.125	1.000	2.062	1.000
CQ--A-PF224-3	600	.22	F, J, K	2.625	1.000	2.562	1.000
Characteristic T 6/				Case dimensions			
CQ--A-TC152-3	200	.0015	G, J, K	.812	.235	.719	.235 2/
CQ--A-TC392-3	200	.0039	G, J, K	.938	.312	.844	.312 2/
CQ--A-TC472-3	200	.0047	G, J, K	.938	.312	.844	.312 2/
CQ--A-TC822-3	200	.0082	G, J, K	.938	.400	.844	.400
CQ--A-TC103-3	200	.010	G, J, K	.938	.400	.844	.400
CQ--A-TC123-3	200	.012	G, J, K	1.188	.400	1.094	.400
CQ--A-TC153-3	200	.015	G, J, K	1.188	.400	1.094	.400
CQ--A-TC183-3	200	.018	G, J, K	1.312	.400	1.219	.400
CQ--A-TC223-3	200	.022	G, J, K	1.312	.400	1.219	.400
CQ--A-TC273-3	200	.027	G, J, K	1.125	.562	1.062	.562
CQ--A-TC333-3	200	.033	G, J, K	1.125	.562	1.062	.562
CQ--A-TC393-3	200	.039	G, J, K	1.250	.562	1.188	.562
CQ--A-TC473-3	200	.047	G, J, K	1.250	.562	1.188	.562
CQ--A-TC563-3	200	.056	G, J, K	1.500	.562	1.438	.562
CQ--A-TC683-3	200	.068	G, J, K	1.500	.562	1.438	.562
CQ--A-TC823-3	200	.082	G, J, K	1.875	.562	1.812	.562
CQ--A-TC104-3	200	.10	G, J, K	1.875	.562	1.812	.562
CQ--A-TE104-3	400	.10	G, J, K	1.875	.670	1.812	.670
CQ--A-TE124-3	400	.12	G, J, K	2.125	.705	2.062	.750
CQ--A-TE154-3	400	.15	G, J, K	2.125	.705	2.062	.750
CQ--A-TE184-3	400	.18	G, J, K	1.875	1.000	1.812	1.000
CQ--A-TE224-3	400	.22	G, J, K	1.875	1.000	1.812	1.000
CQ--A-TE274-3	400	.27	G, J, K	2.375	1.000	2.312	1.000
CQ--A-TE334-3	400	.33	G, J, K	2.375	1.000	2.312	1.000
CQ--A-TF102-3	600	.0010	G, J, K	.812	.235	.719	.235 2/
CQ--A-TF122-3	600	.0012	G, J, K	.938	.312	.814	.312 2/
CQ--A-TF152-3	600	.0015	G, J, K	.938	.312	.814	.312 2/
CQ--A-TF182-3	600	.0018	G, J, K	.938	.312	.814	.312 2/
CQ--A-TF222-3	600	.0022	G, J, K	.938	.312	.814	.312 2/

See footnotes at end of table.

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TABLE III. Style CQ08, CQ09, CQ12, and CQ13 capacitors - Continued.

Characteristic T 6/				Case dimensions			
Part or identifying number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Styles CQ08, CQ12, and CQ13 only (For style CQ09 see 3/ 4/)			
				Circuit 1		Circuit 3	
				L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQ--A-TF272-3	600	.0027	G, J, K	.938	.312	.814	.312 2/
CQ--A-TF332-3	600	.0033	G, J, K	.938	.312	.814	.312 2/
CQ--A-TF392-3	600	.0039	G, J, K	.938	.400	.844	.400
CQ--A-TF472-3	600	.0047	G, J, K	.938	.400	.844	.400
CQ--A-TF562-3	600	.0056	G, J, K	.938	.400	.844	.400
CQ--A-TF682-3	600	.0068	G, J, K	.938	.400	.844	.400
CQ--A-TF822-3	600	.0082	G, J, K	1.188	.400	1.094	.400
CQ--A-TF103-3	600	.010	G, J, K	1.188	.400	1.094	.400
CQ--A-TF123-3	600	.012	G, J, K	1.312	.400	1.219	.400
CQ--A-TF153-3	600	.015	G, J, K	1.312	.400	1.219	.400
CQ--A-TF183-3	600	.018	G, J, K	1.125	.562	1.062	.562
CQ--A-TF223-3	600	.022	G, J, K	1.125	.562	1.062	.562
CQ--A-TF273-3	600	.027	G, J, K	1.250	.562	1.188	.562
CQ--A-TF333-3	600	.033	G, J, K	1.250	.562	1.188	.562
CQ--A-TF393-3	600	.039	G, J, K	1.500	.562	1.438	.562
CQ--A-TF473-3	600	.047	G, J, K	1.500	.562	1.438	.562
CQ--A-TF563-3	600	.056	G, J, K	1.875	.562	1.812	.562
CQ--A-TF683-3	600	.068	G, J, K	1.875	.562	1.812	.562
CQ--A-TF823-3	600	.082	G, J, K	1.875	1.000	1.812	1.000
CQ--A-TF104-3	600	.10	G, J, K	1.875	1.000	1.812	1.000
CQ--A-TF124-3	600	.12	G, J, K	1.875	1.000	2.312	1.000
CQ--A-TF154-3	600	.15	G, J, K	1.875	1.000	2.312	1.000

1/ Complete part number shall include additional symbols to complete style designation, and to indicate circuit, and capacitance tolerance, as applicable.
2/ The minimum D dimension for style CQ13 shall be .400 inch.
3/ For insulated style CQ09 in characteristics K, M, and P add .062 +.062, -.093 inch and .032 +.015, -.037 inch to the nominal L and D dimensions respectively. In no case shall the bare case dimensions of the insulated style be less than the uninsulated minimum.
4/ For insulated style CQ09 in characteristic T add .062 +.062, -.093 inch and .062 +.015, -.068 inch to the nominal L and D dimensions respectively. In no case shall the bare case dimensions of the insulated style be less than the uninsulated minimum.
5/ These units are not available in style CQ13 due to possible arcing with 1,000 volts impressed across the smaller screw necks.
6/ Styles CQ08, CQ09, CQ12 and CQ13 replace and are directly interchangeable with former styles CQ04, CQ05, CQ10, and CQ11 respectively of the same characteristic, voltage, and tolerance.

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TABLE IV. Millimeter equivalents of decimal inches.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.005	0.13	.688	17.48	1.125	28.58	1.812	46.02
.015	0.38	.718	18.24	1.156	29.36	1.875	47.63
.031	0.79	.719	18.26	1.188	30.18	1.906	48.41
.175	4.45	.749	19.02	1.219	30.96	2.031	51.59
.195	4.95	.750	19.05	1.250	31.75	2.062	52.37
.235	5.97	.781	19.84	1.281	32.54	2.125	53.98
.312	7.92	.812	20.62	1.312	33.32	2.156	54.76
.400	10.16	.844	21.44	1.375	34.93	2.281	57.94
.406	10.31	.875	22.23	1.406	35.71	2.312	58.72
.468	11.89	.906	23.01	1.438	36.53	2.375	60.33
.500	12.70	.938	23.83	1.469	37.31	2.531	64.29
.562	14.27	.969	24.61	1.500	38.10	2.562	65.07
.625	15.88	1.000	25.40	1.531	38.89	2.625	66.68
.656	16.66	1.031	26.10	1.562	39.67	2.656	67.46
.670	17.02	1.062	26.97	1.625	41.28		
.687	17.45	1.094	27.79	1.781	45.24		

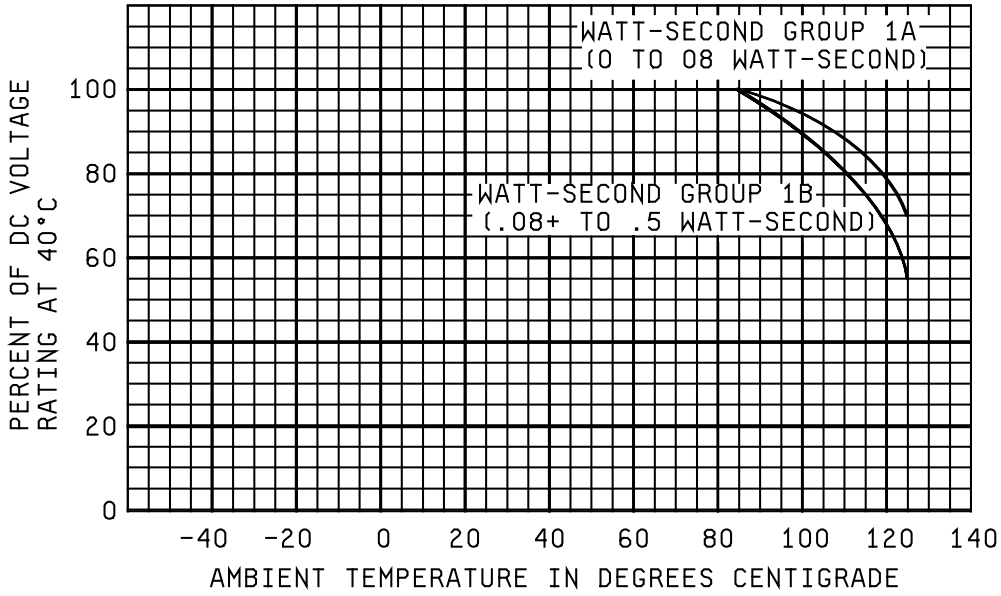


FIGURE 2. Voltage derating for high ambient test temperature (characteristic K).

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TABLE V. Substitutability data.

Superseding part number 1/ MIL-PRF-19978/9,10,11 Characteristic K CQR09, CQR12, CQR13	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-KC392-3-	0001-	0002-	0053-	0054-	0105-	0106-	0157-	0158-	CQ--A-KC392-3
CQR--A-KC472-3-	0003-	0004-	0055-	0056-	0107-	0108-	0159-	0160-	CQ--A-KC472-3
CQR--A-KC562-3-	0005-	0006-	0057-	0058-	0109-	0110-	0161-	0162-	CQ--A-KC562-3
CQR--A-KC682-3-	0007-	0008-	0059-	0060-	0111-	0112-	0163-	0164-	CQ--A-KC682-3
CQR--A-KC183-3-	0009-	0010-	0061-	0062-	0113-	0114-	0165-	0166-	CQ--A-KC183-3
CQR--A-KC223-3-	0011-	0012-	0063-	0064-	0115-	0116-	0167-	0168-	CQ--A-KC223-3
CQR--A-KC273-3-	0013-	0014-	0065-	0066-	0117-	0118-	0169-	0170-	CQ--A-KC273-3
CQR--A-KC333-3-	0015-	0016-	0067-	0068-	0119-	0120-	0171-	0172-	CQ--A-KC333-3
CQR--A-KC393-3-	0017-	0018-	0069-	0070-	0121-	0122-	0173-	0174-	CQ--A-KC393-3
CQR--A-KC473-3-	0019-	0020-	0071-	0072-	0123-	0124-	0175-	0176-	CQ--A-KC473-3
CQR--A-KC563-3-	0021-	0022-	0073-	0074-	0125-	0126-	0177-	0178-	CQ--A-KC563-3
CQR--A-KC683-3-	0023-	0024-	0075-	0076-	0127-	0128-	0179-	0180-	CQ--A-KC683-3
CQR--A-KC823-3-	0025-	0026-	0077-	0078-	0129-	0130-	0181-	0182-	CQ--A-KC823-3
CQR--A-KC104-3-	0027-	0028-	0079-	0080-	0131-	0132-	0183-	0184-	CQ--A-KC104-3
CQR--A-KC124-3-	0029-	0030-	0081-	0082-	0133-	0134-	0185-	0186-	CQ--A-KC124-3
CQR--A-KC154-3-	0031-	0032-	0083-	0084-	0135-	0136-	0187-	0188-	CQ--A-KC154-3
CQR--A-KC184-3-	0033-	0034-	0085-	0086-	0137-	0138-	0189-	0190-	CQ--A-KC184-3
CQR--A-KC224-3-	0035-	0036-	0087-	0088-	0139-	0140-	0191-	0192-	CQ--A-KC224-3
CQR--A-KC274-3-	0037-	0038-	0089-	0090-	0141-	0142-	0193-	0194-	CQ--A-KC274-3
CQR--A-KC334-3-	0039-	0040-	0091-	0092-	0143-	0144-	0195-	0196-	CQ--A-KC334-3
CQR--A-KC394-3-	0041-	0042-	0093-	0094-	0145-	0146-	0197-	0198-	CQ--A-KC394-3
CQR--A-KC474-3-	0043-	0044-	0095-	0096-	0147-	0148-	0199-	0200-	CQ--A-KC474-3
CQR--A-KC564-3-	0045-	0046-	0097-	0098-	0149-	0150-	0201-	0202-	CQ--A-KC564-3
CQR--A-KC684-3-	0047-	0048-	0099-	0100-	0151-	0152-	0203-	0204-	CQ--A-KC684-3
CQR--A-KC824-3-	0049-	0050-	0101-	0102-	0153-	0154-	0205-	0206-	CQ--A-KC824-3
CQR--A-KC105-3-	0051-	0052-	0103-	0104-	0155-	0156-	0207-	0208-	CQ--A-KC105-3
CQR--A-KE272-3-	0301-	0302-	0341-	0342-	0381-	0382-	0421-	0422-	CQ--A-KE272-3
CQR--A-KE332-3-	0303-	0304-	0343-	0344-	0383-	0384-	0423-	0424-	CQ--A-KE332-3
CQR--A-KE123-3-	0305-	0306-	0345-	0346-	0385-	0386-	0425-	0426-	CQ--A-KE123-3
CQR--A-KE153-3-	0307-	0308-	0347-	0348-	0387-	0388-	0427-	0428-	CQ--A-KE153-3
CQR--A-KE273-3-	0309-	0310-	0349-	0350-	0389-	0390-	0429-	0430-	CQ--A-KE273-3
CQR--A-KE333-3-	0311-	0312-	0351-	0352-	0391-	0392-	0431-	0432-	CQ--A-KE333-3
CQR--A-KE393-3-	0313-	0314-	0353-	0354-	0393-	0394-	0433-	0434-	CQ--A-KE393-3
CQR--A-KE473-3-	0315-	0316-	0355-	0356-	0395-	0396-	0435-	0436-	CQ--A-KE473-3
CQR--A-KE563-3-	0317-	0318-	0357-	0358-	0397-	0398-	0437-	0438-	CQ--A-KE563-3
CQR--A-KE683-3-	0319-	0320-	0359-	0360-	0399-	0400-	0439-	0440-	CQ--A-KE683-3
CQR--A-KE823-3-	0321-	0322-	0361-	0362-	0401-	0402-	0441-	0442-	CQ--A-KE823-3
CQR--A-KE104-3-	0323-	0324-	0363-	0364-	0403-	0404-	0443-	0444-	CQ--A-KE104-3
CQR--A-KE124-3-	0325-	0326-	0365-	0366-	0405-	0406-	0445-	0446-	CQ--A-KE124-3
CQR--A-KE154-3-	0327-	0328-	0367-	0368-	0407-	0408-	0447-	0448-	CQ--A-KE154-3
CQR--A-KE184-3-	0329-	0330-	0369-	0370-	0409-	0410-	0449-	0450-	CQ--A-KE184-3
CQR--A-KE224-3-	0331-	0332-	0371-	0372-	0411-	0412-	0451-	0452-	CQ--A-KE224-3
CQR--A-KE274-3-	0333-	0334-	0373-	0374-	0413-	0414-	0453-	0454-	CQ--A-KE274-3
CQR--A-KE334-3-	0335-	0336-	0375-	0376-	0415-	0416-	0455-	0456-	CQ--A-KE334-3
CQR--A-KE394-3-	0337-	0338-	0377-	0378-	0417-	0418-	0457-	0458-	CQ--A-KE394-3
CQR--A-KE474-3-	0339-	0340-	0379-	0380-	0419-	0420-	0459-	0460-	CQ--A-KE474-3
CQR--A-KF102-3-	0501-	0502-	0567-	0568-	0633-	0634-	0699-	0700-	CQ--A-KF102-3
CQR--A-KF122-3-	0503-	0504-	0569-	0570-	0635-	0636-	0701-	0702-	CQ--A-KF122-3
CQR--A-KF152-3-	0505-	0506-	0571-	0572-	0637-	0638-	0703-	0704-	CQ--A-KF152-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-PRF-19978/9,10,11 Characteristic K CQR09, CQR12, CQR13	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-KF182-3-	0507-	0508-	0573-	0574-	0639-	0640-	0705-	0706-	CQ--A-KF182-3
CQR--A-KF222-3-	0509-	0510-	0575-	0576-	0641-	0642-	0707-	0708-	CQ--A-KF222-3
CQR--A-KF272-3-	0511-	0512-	0577-	0578-	0643-	0644-	0709-	0710-	CQ--A-KF272-3
CQR--A-KF332-3-	0513-	0514-	0579-	0580-	0645-	0646-	0711-	0712-	CQ--A-KF332-3
CQR--A-KF392-3-	0515-	0516-	0581-	0582-	0647-	0648-	0713-	0714-	CQ--A-KF392-3
CQR--A-KF472-3-	0517-	0518-	0583-	0584-	0649-	0650-	0715-	0716-	CQ--A-KF472-3
CQR--A-KF562-3-	0519-	0520-	0585-	0586-	0651-	0652-	0717-	0718-	CQ--A-KF562-3
CQR--A-KF682-3-	0521-	0522-	0587-	0588-	0653-	0654-	0719-	0720-	CQ--A-KF682-3
CQR--A-KF822-3-	0523-	0524-	0589-	0590-	0655-	0656-	0721-	0722-	CQ--A-KF822-3
CQR--A-KF103-3-	0525-	0526-	0591-	0592-	0657-	0658-	0723-	0724-	CQ--A-KF103-3
CQR--A-KF123-3-	0527-	0528-	0593-	0594-	0659-	0660-	0725-	0726-	CQ--A-KF123-3
CQR--A-KF153-3-	0529-	0530-	0595-	0596-	0661-	0662-	0727-	0728-	CQ--A-KF153-3
CQR--A-KF183-3-	0531-	0532-	0597-	0598-	0663-	0664-	0729-	0730-	CQ--A-KF183-3
CQR--A-KF223-3-	0533-	0534-	0599-	0600-	0665-	0666-	0731-	0732-	CQ--A-KF223-3
CQR--A-KF273-3-	0535-	0536-	0601-	0602-	0667-	0668-	0733-	0734-	CQ--A-KF273-3
CQR--A-KF333-3-	0537-	0538-	0603-	0604-	0669-	0670-	0735-	0736-	CQ--A-KF333-3
CQR--A-KF393-3-	0539-	0540-	0605-	0606-	0671-	0672-	0737-	0738-	CQ--A-KF393-3
CQR--A-KF473-3-	0541-	0542-	0607-	0608-	0673-	0674-	0739-	0740-	CQ--A-KF473-3
CQR--A-KF563-3-	0543-	0544-	0609-	0610-	0675-	0676-	0741-	0742-	CQ--A-KF563-3
CQR--A-KF683-3-	0545-	0546-	0611-	0612-	0677-	0678-	0743-	0744-	CQ--A-KF683-3
CQR--A-KF823-3-	0547-	0548-	0613-	0614-	0679-	0680-	0745-	0746-	CQ--A-KF823-3
CQR--A-KF104-3-	0549-	0550-	0615-	0616-	0681-	0682-	0747-	0748-	CQ--A-KF104-3
CQR--A-KF124-3-	0551-	0552-	0617-	0618-	0683-	0684-	0749-	0750-	CQ--A-KF124-3
CQR--A-KF154-3-	0553-	0554-	0619-	0620-	0685-	0686-	0751-	0752-	CQ--A-KF154-3
CQR--A-KF184-3-	0555-	0556-	0621-	0622-	0687-	0688-	0753-	0754-	CQ--A-KF184-3
CQR--A-KF224-3-	0557-	0558-	0623-	0624-	0689-	0690-	0755-	0756-	CQ--A-KF224-3
CQR--A-KF274-3-	0559-	0560-	0625-	0626-	0691-	0692-	0757-	0758-	CQ--A-KF274-3
CQR--A-KF334-3-	0561-	0562-	0627-	0628-	0693-	0694-	0759-	0760-	CQ--A-KF334-3
CQR--A-KF394-3-	0563-	0564-	0629-	0630-	0695-	0696-	0761-	0762-	CQ--A-KF394-3
CQR--A-KF474-3-	0565-	0566-	0631-	0632-	0697-	0698-	0763-	0764-	CQ--A-KF474-3
CQR--A-KG102-3-	0801-	0802-	0867-	0868-	0933-	0934-	0999-	1000-	CQ--A-KG102-3
CQR--A-KG122-3-	0803-	0804-	0869-	0870-	0935-	0936-	1001-	1002-	CQ--A-KG122-3
CQR--A-KG152-3-	0805-	0806-	0871-	0872-	0937-	0938-	1003-	1004-	CQ--A-KG152-3
CQR--A-KG182-3-	0807-	0808-	0873-	0874-	0939-	0940-	1005-	1006-	CQ--A-KG182-3
CQR--A-KG222-3-	0809-	0810-	0875-	0876-	0941-	0942-	1007-	1008-	CQ--A-KG222-3
CQR--A-KG272-3-	0811-	0812-	0877-	0878-	0943-	0944-	1009-	1010-	CQ--A-KG272-3
CQR--A-KG332-3-	0813-	0814-	0879-	0880-	0945-	0946-	1011-	1012-	CQ--A-KG332-3
CQR--A-KG392-3-	0815-	0816-	0881-	0882-	0947-	0948-	1013-	1014-	CQ--A-KG392-3
CQR--A-KG472-3-	0817-	0818-	0883-	0884-	0949-	0950-	1015-	1016-	CQ--A-KG472-3
CQR--A-KG562-3-	0819-	0820-	0885-	0886-	0951-	0952-	1017-	1018-	CQ--A-KG562-3
CQR--A-KG682-3-	0821-	0822-	0887-	0888-	0953-	0954-	1019-	1020-	CQ--A-KG682-3
CQR--A-KG822-3-	0823-	0824-	0889-	0890-	0955-	0956-	1021-	1022-	CQ--A-KG822-3
CQR--A-KG103-3-	0825-	0826-	0891-	0892-	0957-	0958-	1023-	1024-	CQ--A-KG103-3
CQR--A-KG123-3-	0827-	0828-	0893-	0894-	0959-	0960-	1025-	1026-	CQ--A-KG123-3
CQR--A-KG153-3-	0829-	0830-	0895-	0896-	0961-	0962-	1027-	1028-	CQ--A-KG153-3
CQR--A-KG183-3-	0831-	0832-	0897-	0898-	0963-	0964-	1029-	1030-	CQ--A-KG183-3
CQR--A-KG223-3-	0833-	0834-	0899-	0900-	0965-	0966-	1031-	1032-	CQ--A-KG223-3
CQR--A-KG273-3-	0835-	0836-	0901-	0902-	0967-	0968-	1033-	1034-	CQ--A-KG273-3
CQR--A-KG333-3-	0837-	0838-	0903-	0904-	0969-	0970-	1035-	1036-	CQ--A-KG333-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-PRF-19978/9,10,11 Characteristic K CQR09, CQR12, CQR13	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-KG393-3-	0839-	0840-	0905-	0906-	0971-	0972-	1037-	1038-	CQ--A-KG393-3
CQR--A-KG473-3-	0841-	0842-	0907-	0908-	0973-	0974-	1039-	1040-	CQ--A-KG473-3
CQR--A-KG563-3-	0843-	0844-	0909-	0910-	0975-	0976-	1041-	1042-	CQ--A-KG563-3
CQR--A-KG683-3-	0845-	0846-	0911-	0912-	0977-	0978-	1043-	1044-	CQ--A-KG683-3
CQR--A-KG823-3-	0847-	0848-	0913-	0914-	0979-	0980-	1045-	1046-	CQ--A-KG823-3
CQR--A-KG104-3-	0849-	0850-	0915-	0916-	0981-	0982-	1047-	1048-	CQ--A-KG104-3
CQR--A-KG124-3-	0851-	0852-	0917-	0918-	0983-	0984-	1049-	1050-	CQ--A-KG124-3
CQR--A-KG154-3-	0853-	0854-	0919-	0920-	0985-	0986-	1051-	1052-	CQ--A-KG154-3
CQR--A-KG184-3-	0855-	0856-	0921-	0922-	0987-	0988-	1053-	1054-	CQ--A-KG184-3
CQR--A-KG224-3-	0857-	0858-	0923-	0924-	0989-	0990-	1055-	1056-	CQ--A-KG224-3
CQR--A-KG274-3-	0859-	0860-	0925-	0926-	0991-	0992-	1057-	1058-	CQ--A-KG274-3
CQR--A-KG334-3-	0861-	0862-	0927-	0928-	0993-	0994-	1059-	1060-	CQ--A-KG334-3
CQR--A-KG394-3-	0863-	0864-	0929-	0930-	0995-	0996-	1061-	1062-	CQ--A-KG394-3
CQR--A-KG474-3-	0865-	0866-	0931-	0932-	0997-	0998-	1063-	1064-	CQ--A-KG474-3
Superseding part number 1/ MIL-PRF-19978/13,14,15 Characteristic M CQR29, CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-MZ155-3-	1101-	1102-	1113-	1114-	1125-	1126-	1137-	1138-	CQ--A-MZ155-3-
CQR--A-MZ225-3-	1103-	1104-	1115-	1116-	1127-	1128-	1139-	1140-	CQ--A-MZ225-3-
CQR--A-MZ335-3-	1105-	1106-	1117-	1118-	1129-	1130-	1141-	1142-	CQ--A-MZ335-3-
CQR--A-MZ475-3-	1107-	1108-	1119-	1120-	1131-	1132-	1143-	1144-	CQ--A-MZ475-3-
CQR--A-MZ685-3-	1109-	1110-	1121-	1122-	1133-	1134-	1145-	1146-	CQ--A-MZ685-3-
CQR--A-MZ106-3-	1111-	1112-	1123-	1124-	1135-	1136-	1147-	1148-	CQ--A-MZ106-3-
CQR--A-MA102-3-	1201-	1202-	1273-	1274-	1345-	1346-	1237-	1238-	CQ--A-MA102-3-
CQR--A-MA122-3-	1203-	1204-	1275-	1276-	1347-	1348-	1239-	1240-	CQ--A-MA122-3-
CQR--A-MA152-3-	1205-	1206-	1277-	1278-	1349-	1350-	1241-	1242-	CQ--A-MA152-3-
CQR--A-MA182-3-	1207-	1208-	1279-	1280-	1351-	1352-	1243-	1244-	CQ--A-MA182-3-
CQR--A-MA222-3-	1209-	1210-	1281-	1282-	1353-	1354-	1245-	1246-	CQ--A-MA222-3-
CQR--A-MA272-3-	1211-	1212-	1283-	1284-	1355-	1356-	1427-	1428-	CQ--A-MA272-3-
CQR--A-MA332-3-	1213-	1214-	1285-	1286-	1357-	1358-	1429-	1430-	CQ--A-MA332-3-
CQR--A-MA392-3-	1215-	1216-	1287-	1288-	1359-	1360-	1431-	1432-	CQ--A-MA392-3-
CQR--A-MA472-3-	1217-	1218-	1289-	1290-	1361-	1362-	1433-	1434-	CQ--A-MA472-3-
CQR--A-MA562-3-	1219-	1220-	1291-	1292-	1363-	1364-	1435-	1436-	CQ--A-MA562-3-
CQR--A-MA682-3-	1221-	1222-	1293-	1294-	1365-	1366-	1437-	1438-	CQ--A-MA682-3-
CQR--A-MA822-3-	1223-	1224-	1295-	1296-	1367-	1368-	1439-	1440-	CQ--A-MA822-3-
CQR--A-MA103-3-	1225-	1226-	1297-	1298-	1369-	1370-	1441-	1442-	CQ--A-MA103-3-
CQR--A-MA123-3-	1227-	1228-	1299-	1300-	1371-	1372-	1443-	1444-	CQ--A-MA123-3-
CQR--A-MA153-3-	1229-	1230-	1301-	1302-	1373-	1374-	1445-	1446-	CQ--A-MA153-3-
CQR--A-MA183-3-	1231-	1232-	1303-	1304-	1375-	1376-	1447-	1448-	CQ--A-MA183-3-
CQR--A-MA223-3-	1233-	1234-	1305-	1306-	1377-	1378-	1449-	1450-	CQ--A-MA223-3-
CQR--A-MA273-3-	1235-	1236-	1307-	1308-	1379-	1380-	1451-	1452-	CQ--A-MA273-3-
CQR--A-MA333-3-	1237-	1238-	1309-	1310-	1381-	1382-	1453-	1454-	CQ--A-MA333-3-
CQR--A-MA393-3-	1239-	1240-	1311-	1312-	1383-	1384-	1455-	1456-	CQ--A-MA393-3-
CQR--A-MA473-3-	1241-	1242-	1313-	1314-	1385-	1386-	1457-	1458-	CQ--A-MA473-3
CQR--A-MA563-3-	1243-	1244-	1315-	1316-	1387-	1388-	1459-	1460-	CQ--A-MA563-3
CQR--A-MA683-3-	1245-	1246-	1317-	1318-	1389-	1390-	1461-	1462-	CQ--A-MA683-3
CQR--A-MA124-3-	1247-	1248-	1319-	1320-	1391-	1392-	1463-	1464-	CQ--A-MA124-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-PRF-19978/13, 14, 15 Characteristic M CQR29, CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-MA154-3-	1249-	1250-	1321-	1322-	1393-	1394-	1465-	1466-	CQ--A-MA154-3
CQR--A-MA184-3-	1251-	1252-	1323-	1324-	1395-	1396-	1467-	1468-	CQ--A-MA184-3
CQR--A-MA224-3-	1253-	1254-	1325-	1326-	1397-	1398-	1469-	1470-	CQ--A-MA224-3
CQR--A-MA274-3-	1255-	1256-	1327-	1328-	1399-	1400-	1471-	1472-	CQ--A-MA274-3
CQR--A-MA334-3-	1257-	1258-	1329-	1330-	1401-	1402-	1473-	1474-	CQ--A-MA334-3
CQR--A-MA394-3-	1259-	1260-	1331-	1332-	1403-	1404-	1475-	1476-	CQ--A-MA394-3
CQR--A-MA474-3-	1261-	1262-	1333-	1334-	1405-	1406-	1477-	1478-	CQ--A-MA474-3
CQR--A-MA564-3-	1263-	1264-	1335-	1336-	1407-	1408-	1479-	1480-	CQ--A-MA564-3
CQR--A-MA684-3-	1265-	1266-	1337-	1338-	1409-	1410-	1481-	1482-	CQ--A-MA684-3
CQR--A-MA824-3-	1267-	1268-	1339-	1340-	1411-	1412-	1483-	1484-	CQ--A-MA824-3
CQR--A-MA105-3-	1269-	1270-	1341-	1342-	1413-	1414-	1485-	1486-	CQ--A-MA105-3
CQR--A-MA125-3-	1271-	1272-	1343-	1344-	1415-	1416-	1487-	1488-	CQ--A-MA125-3
CQR--A-MB125-3-	1501-	1502-	1521-	1522-	1541-	1542-	1561-	1562-	CQ--A-MB125-3
CQR--A-MB155-3-	1503-	1504-	1523-	1524-	1543-	1544-	1563-	1564-	CQ--A-MB155-3
CQR--A-MB185-3-	1505-	1506-	1525-	1526-	1545-	1546-	1565-	1566-	CQ--A-MB185-3
CQR--A-MB225-3-	1507-	1508-	1527-	1528-	1547-	1548-	1567-	1568-	CQ--A-MB225-3
CQR--A-MB275-3-	1509-	1510-	1529-	1530-	1549-	1550-	1569-	1570-	CQ--A-MB275-3
CQR--A-MB335-3-	1511-	1512-	1531-	1532-	1551-	1552-	1571-	1572-	CQ--A-MB335-3
CQR--A-MB395-3-	1513-	1514-	1533-	1534-	1553-	1554-	1573-	1574-	CQ--A-MB395-3
CQR--A-MB475-3-	1515-	1516-	1535-	1536-	1555-	1556-	1575-	1576-	CQ--A-MB475-3
CQR--A-MB565-3-	1517-	1518-	1537-	1538-	1557-	1558-	1577-	1578-	CQ--A-MB565-3
CQR--A-MB685-3-	1519-	1520-	1539-	1540-	1559-	1560-	1579-	1580-	CQ--A-MB685-3
CQR--A-MC102-3-	1601-	1602-	1675-	1676-	1749-	1750-	1823-	1824-	CQ--A-MC102-3
CQR--A-MC122-3-	1603-	1604-	1677-	1678-	1751-	1752-	1825-	1826-	CQ--A-MC122-3
CQR--A-MC152-3-	1605-	1606-	1679-	1680-	1753-	1754-	1827-	1828-	CQ--A-MC152-3
CQR--A-MC182-3-	1607-	1608-	1681-	1682-	1755-	1756-	1829-	1830-	CQ--A-MC182-3
CQR--A-MC222-3-	1609-	1610-	1683-	1684-	1757-	1758-	1831-	1832-	CQ--A-MC222-3
CQR--A-MC272-3-	1611-	1612-	1685-	1686-	1759-	1760-	1833-	1834	CQ--A-MC272-3
CQR--A-MC332-3-	1613-	1614-	1687-	1688-	1761-	1762-	1835-	1836-	CQ--A-MC332-3
CQR--A-MC392-3-	1615-	1616-	1689-	1690-	1763-	1764-	1837-	1838-	CQ--A-MC392-3
CQR--A-MC472-3-	1617-	1618-	1691-	1692-	1765-	1766-	1839-	1840-	CQ--A-MC472-3
CQR--A-MC562-3-	1619-	1620-	1693-	1694-	1767-	1768-	1841-	1842-	CQ--A-MC562-3
CQR--A-MC682-3-	1621-	1622-	1695-	1696-	1769-	1770-	1843-	1844-	CQ--A-MC682-3
CQR--A-MC822-3-	1623-	1624-	1697-	1698-	1771-	1772-	1845-	1846-	CQ--A-MC822-3
CQR--A-MC103-3-	1625-	1626-	1699-	1700-	1773-	1774-	1847-	1848-	CQ--A-MC103-3
CQR--A-MC123-3-	1627-	1628-	1701-	1702-	1775-	1776-	1849-	1850-	CQ--A-MC123-3
CQR--A-MC153-3-	1629-	1630-	1703-	1704-	1777-	1778-	1851-	1852-	CQ--A-MC153-3
CQR--A-MC183-3-	1631-	1632-	1705-	1706-	1779-	1780-	1853-	1854-	CQ--A-MC183-3
CQR--A-MC223-3-	1633-	1634-	1707-	1708-	1781-	1782-	1855-	1856-	CQ--A-MC223-3
CQR--A-MC273-3-	1635-	1636-	1709-	1710-	1783-	1784-	1857-	1858-	CQ--A-MC273-3
CQR--A-MC333-3-	1637-	1638-	1711-	1712-	1785-	1786-	1859-	1860-	CQ--A-MC333-3
CQR--A-MC393-3-	1639-	1640-	1713-	1714-	1787-	1788-	1861-	1862-	CQ--A-MC393-3
CQR--A-MC473-3-	1641-	1642-	1715-	1716-	1789-	1790-	1863-	1864-	CQ--A-MC473-3
CQR--A-MC563-3-	1643-	1644-	1717-	1718-	1791-	1792-	1865-	1866-	CQ--A-MC563-3
CQR--A-MC683-3-	1645-	1646-	1719-	1720-	1793-	1794-	1867-	1868-	CQ--A-MC683-3
CQR--A-MC823-3-	1647-	1648-	1721-	1722-	1795-	1796-	1869-	1870-	CQ--A-MC823-3
CQR--A-MC104-3-	1649-	1650-	1723-	1724-	1797-	1798-	1871-	1872-	CQ--A-MC104-3
CQR--A-MC124-3-	1651-	1652-	1725-	1726-	1799-	1800-	1873-	1874-	CQ--A-MC124-3
CQR--A-MC154-3-	1653-	1654-	1727-	1728-	1801-	1802-	1875-	1876-	CQ--A-MC154-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-PRF-19978/13,14,15 Characteristic M CQR29, CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-MC184-3-	1655-	1656-	1729-	1730-	1803-	1804-	1877-	1878-	CQ--A-MC184-3
CQR--A-MC224-3-	1657-	1658-	1731-	1732-	1805-	1806-	1879-	1880-	CQ--A-MC224-3
CQR--A-MC274-3-	1659-	1660-	1733-	1734-	1807-	1808-	1881-	1882-	CQ--A-MC274-3
CQR--A-MC334-3-	1661-	1662-	1735-	1736-	1809-	1810-	1883-	1884-	CQ--A-MC334-3
CQR--A-MC394-3-	1663-	1664-	1737-	1738-	1811-	1812-	1885-	1886-	CQ--A-MC394-3
CQR--A-MC474-3-	1665-	1666-	1739-	1740-	1813-	1814-	1887-	1888-	CQ--A-MC474-3
CQR--A-MC564-3-	1667-	1668-	1741-	1742-	1815-	1816-	1889-	1890-	CQ--A-MC564-3
CQR--A-MC684-3-	1669-	1670-	1743-	1744-	1817-	1818-	1891-	1892-	CQ--A-MC684-3
CQR--A-MC824-3-	1671-	1672-	1745-	1746-	1819-	1820-	1893-	1894-	CQ--A-MC824-3
CQR--A-MC105-3-	1673-	1674-	1747-	1748-	1821-	1822-	1895-	1896-	CQ--A-MC105-3
CQR--A-ME102-3-	2001-	2002-	2075-	2076-	2149-	2150-	2223-	2224-	CQ--A-ME102-3
CQR--A-ME122-3-	2003-	2004-	2077-	2078-	2151-	2152-	2225-	2226-	CQ--A-ME122-3
CQR--A-ME152-3-	2005-	2006-	2079-	2080-	2153-	2154-	2227-	2228-	CQ--A-ME152-3
CQR--A-ME182-3-	2007-	2008-	2081-	2082-	2155-	2156-	2229-	2230-	CQ--A-ME182-3
CQR--A-ME222-3-	2009-	2010-	2083-	2084-	2157-	2158-	2231-	2232-	CQ--A-ME222-3
CQR--A-ME272-3-	2011-	2012-	2085-	2086-	2159-	2160-	2233-	2234-	CQ--A-ME272-3
CQR--A-ME332-3-	2013-	2014-	2087-	2088-	2161-	2162-	2235-	2236-	CQ--A-ME332-3
CQR--A-ME392-3-	2015-	2016-	2089-	2090-	2163-	2164-	2237-	2238-	CQ--A-ME392-3
CQR--A-ME472-3-	2017-	2018-	2091-	2092-	2165-	2166-	2239-	2240-	CQ--A-ME472-3
CQR--A-ME562-3-	2019-	2020-	2093-	2094-	2167-	2168-	2241-	2242-	CQ--A-ME562-3
CQR--A-ME682-3-	2021-	2022-	2095-	2096-	2169-	2170-	2243-	2244-	CQ--A-ME682-3
CQR--A-ME822-3-	2023-	2024-	2097-	2098-	2171-	2172-	2245-	2246-	CQ--A-ME822-3
CQR--A-ME103-3-	2025-	2026-	2099-	2100-	2173-	2174-	2247-	2248-	CQ--A-ME103-3
CQR--A-ME123-3-	2027-	2028-	2101-	2102-	2175-	2176-	2249-	2250-	CQ--A-ME123-3
CQR--A-ME153-3-	2029-	2030-	2103-	2104-	2177-	2178-	2251-	2252-	CQ--A-ME153-3
CQR--A-ME183-3-	2031-	2032-	2105-	2106-	2179-	2180-	2253-	2254-	CQ--A-ME183-3
CQR--A-ME223-3-	2033-	2034-	2107-	2108-	2181-	2182-	2255-	2256-	CQ--A-ME223-3
CQR--A-ME273-3-	2035-	2036-	2109-	2110-	2183-	2184-	2257-	2258-	CQ--A-ME273-3
CQR--A-ME333-3-	2037-	2038-	2111-	2112-	2185-	2186-	2259-	2260-	CQ--A-ME333-3
CQR--A-ME393-3-	2039-	2040-	2113-	2114-	2187-	2188-	2261-	2262-	CQ--A-ME393-3
CQR--A-ME473-3-	2041-	2042-	2115-	2116-	2189-	2190-	2263-	2264-	CQ--A-ME473-3
CQR--A-ME563-3-	2043-	2044-	2117-	2118-	2191-	2192-	2265-	2266-	CQ--A-ME563-3
CQR--A-ME683-3-	2045-	2046-	2119-	2120-	2193-	2194-	2267-	2268-	CQ--A-ME683-3
CQR--A-ME823-3-	2047-	2048-	2121-	2122-	2195-	2196-	2269-	2270-	CQ--A-ME823-3
CQR--A-ME104-3-	2049-	2050-	2123-	2124-	2197-	2198-	2271-	2272-	CQ--A-ME104-3
CQR--A-ME124-3-	2051-	2052-	2125-	2126-	2199-	2200-	2273-	2274-	CQ--A-ME124-3
CQR--A-ME154-3-	2053-	2054-	2127-	2128-	2201-	2202-	2275-	2276-	CQ--A-ME154-3
CQR--A-ME184-3-	2055-	2056-	2129-	2130-	2203-	2204-	2277-	2278-	CQ--A-ME184-3
CQR--A-ME224-3-	2057-	2058-	2131-	2132-	2205-	2206-	2279-	2280-	CQ--A-ME224-3
CQR--A-ME274-3-	2059-	2060-	2133-	2134-	2207-	2208-	2281-	2282-	CQ--A-ME274-3
CQR--A-ME334-3-	2061-	2062-	2135-	2136-	2209-	2210-	2283-	2284-	CQ--A-ME334-3
CQR--A-ME394-3-	2063-	2064-	2137-	2138-	2211-	2212-	2285-	2286-	CQ--A-ME394-3
CQR--A-ME474-3-	2065-	2066-	2139-	2140-	2213-	2214-	2287-	2288-	CQ--A-ME474-3
CQR--A-ME564-3-	2067-	2068-	2141-	2142-	2215-	2216-	2289-	2290-	CQ--A-ME564-3
CQR--A-ME684-3-	2069-	2070-	2143-	2144-	2217-	2218-	2291-	2292-	CQ--A-ME684-3
CQR--A-ME824-3-	2071-	2072-	2145-	2146-	2219-	2220-	2293-	2294-	CQ--A-ME824-3
CQR--A-ME105-3-	2073-	2074-	2147-	2148-	2221-	2222-	2295-	2296-	CQ--A-ME105-3
CQR--A-MF102-3-	2401-	2402-	2471-	2472-	2541-	2542-	2611-	2612-	CQ--A-MF102-3
CQR--A-MF122-3-	2403-	2404-	2473-	2474-	2543-	2544-	2613-	2614-	CQ--A-MF122-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-PRF-19978/13,14,15 Characteristic M CQR29, CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-MF152-3-	2405-	2406-	2475-	2476-	2545-	2546-	2615-	2616-	CQ--A-MF152-3
CQR--A-MF182-3-	2407-	2408-	2477-	2478-	2547-	2548-	2617-	2618-	CQ--A-MF182-3
CQR--A-MF222-3-	2409-	2410-	2479-	2480-	2549-	2550-	2619-	2620-	CQ--A-MF222-3
CQR--A-MF272-3-	2411-	2412-	2481-	2482-	2551-	2552-	2621-	2622-	CQ--A-MF272-3
CQR--A-MF332-3-	2413-	2414-	2483-	2484-	2553-	2554-	2623-	2624-	CQ--A-MF332-3
CQR--A-MF392-3-	2415-	2416-	2485-	2486-	2555-	2556-	2625-	2626-	CQ--A-MF392-3
CQR--A-MF472-3-	2417-	2418-	2487-	2488-	2557-	2558-	2627-	2628-	CQ--A-MF472-3
CQR--A-MF562-3-	2419-	2420-	2489-	2490-	2559-	2560-	2629-	2630-	CQ--A-MF562-3
CQR--A-MF682-3-	2421-	2422-	2491-	2492-	2561-	2562-	2631-	2632-	CQ--A-MF682-3
CQR--A-MF822-3-	2423-	2424-	2493-	2494-	2563-	2564-	2633-	2634-	CQ--A-MF822-3
CQR--A-MF103-3-	2425-	2426-	2495-	2496-	2565-	2566-	2635-	2636-	CQ--A-MF103-3
CQR--A-MF123-3-	2427-	2428-	2497-	2498-	2567-	2568-	2637-	2638-	CQ--A-MF123-3
CQR--A-MF153-3-	2429-	2430-	2499-	2500-	2569-	2570-	2639-	2640-	CQ--A-MF153-3
CQR--A-MF183-3-	2431-	2432-	2501-	2502-	2571-	2572-	2641-	2642-	CQ--A-MF183-3
CQR--A-MF223-3-	2433-	2434-	2503-	2504-	2573-	2574-	2643-	2644-	CQ--A-MF223-3
CQR--A-MF273-3-	2435-	2436-	2505-	2506-	2575-	2576-	2645-	2646-	CQ--A-MF273-3
CQR--A-MF333-3-	2437-	2438-	2507-	2508-	2577-	2578-	2647-	2648-	CQ--A-MF333-3
CQR--A-MF393-3-	2439-	2440-	2509-	2510-	2579-	2580-	2649-	2650-	CQ--A-MF393-3
CQR--A-MF473-3-	2441-	2442-	2511-	2512-	2581-	2582-	2651-	2652-	CQ--A-MF473-3
CQR--A-MF563-3-	2443-	2444-	2513-	2514-	2583-	2584-	2653-	2654-	CQ--A-MF563-3
CQR--A-MF683-3-	2445-	2446-	2515-	2516-	2585-	2586-	2655-	2656-	CQ--A-MF683-3
CQR--A-MF823-3-	2447-	2448-	2517-	2518-	2587-	2588-	2657-	2658-	CQ--A-MF823-3
CQR--A-MF104-3-	2449-	2450-	2519-	2520-	2589-	2590-	2659-	2660-	CQ--A-MF104-3
CQR--A-MF124-3-	2451-	2452-	2521-	2522-	2591-	2592-	2661-	2662-	CQ--A-MF124-3
CQR--A-MF154-3-	2453-	2454-	2523-	2524-	2593-	2594-	2663-	2664-	CQ--A-MF154-3
CQR--A-MF184-3-	2455-	2456-	2525-	2526-	2595-	2596-	2665-	2666-	CQ--A-MF184-3
CQR--A-MF224-3-	2457-	2458-	2527-	2528-	2597-	2598-	2667-	2668-	CQ--A-MF224-3
CQR--A-MF274-3-	2459-	2460-	2529-	2530-	2599-	2600-	2669-	2670-	CQ--A-MF274-3
CQR--A-MF334-3-	2461-	2462-	2531-	2532-	2601-	2602-	2671-	2672-	CQ--A-MF334-3
CQR--A-MF394-3-	2463-	2464-	2533-	2534-	2603-	2604-	2673-	2674-	CQ--A-MF394-3
CQR--A-MF474-3-	2465-	2466-	2535-	2536-	2605-	2606-	2675-	2676-	CQ--A-MF474-3
CQR--A-MF564-3-	2467-	2468-	2537-	2538-	2607-	2608-	2677-	2678-	CQ--A-MF564-3
CQR--A-MF684-3-	2469-	2470-	2539-	2540-	2609-	2610-	2679-	2680-	CQ--A-MF684-3
CQR--A-MG102-3-	2701-	2702-	2757-	2758-	2813-	2814-	2869-	2870-	CQ--A-MG102-3
CQR--A-MG122-3-	2703-	2704-	2759-	2760-	2815-	2816-	2871-	2872-	CQ--A-MG122-3
CQR--A-MG152-3-	2705-	2706-	2761-	2762-	2817-	2818-	2873-	2874-	CQ--A-MG152-3
CQR--A-MG182-3-	2707-	2708-	2763-	2764-	2819-	2820-	2875-	2876-	CQ--A-MG182-3
CQR--A-MG222-3-	2709-	2710-	2765-	2766-	2821-	2822-	2877-	2878-	CQ--A-MG222-3
CQR--A-MG272-3-	2711-	2712-	2767-	2768-	2823-	2824-	2879-	2880-	CQ--A-MG272-3
CQR--A-MG332-3-	2713-	2714-	2769-	2770-	2825-	2826-	2881-	2882-	CQ--A-MG332-3
CQR--A-MG392-3-	2715-	2716-	2771-	2772-	2827-	2828-	2883-	2884-	CQ--A-MG392-3
CQR--A-MG472-3-	2717-	2718-	2773-	2774-	2829-	2830-	2885-	2886-	CQ--A-MG472-3
CQR--A-MG562-3-	2719-	2720-	2775-	2776-	2831-	2832-	2887-	2888-	CQ--A-MG562-3
CQR--A-MG682-3-	2721-	2722-	2777-	2778-	2833-	2834-	2889-	2890-	CQ--A-MG682-3
CQR--A-MG822-3-	2723-	2724-	2779-	2780-	2835-	2836-	2891-	2892-	CQ--A-MG822-3
CQR--A-MG103-3-	2725-	2726-	2781-	2782-	2837-	2838-	2893-	2894-	CQ--A-MG103-3
CQR--A-MG123-3-	2727-	2728-	2783-	2784-	2839-	2840-	2895-	2896-	CQ--A-MG123-3
CQR--A-MG183-3-	2729-	2730-	2785-	2786-	2841-	2842-	2897-	2898-	CQ--A-MG183-3
CQR--A-MG223-3-	2731-	2732-	2787-	2788-	2843-	2844-	2899-	2900-	CQ--A-MG223-3

See footnotes at end of table.

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TABLE V. Substitutability data - Continued.

Superseding part number 1/ MIL-PRF-19978/13,14,15 Characteristic M CQR29, CQR32, CQR33	Superseded dash number of MIL-C-19978/1A 2/								Superseded part number 3/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
	Style CQ08		Style CQ09		Style CQ12		Style CQ13		
	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
CQR--A-MG273-3-	2733-	2734-	2789-	2790-	2845-	2846-	2901-	2902-	CQ--A-MG273-3
CQR--A-MG333-3-	2735-	2736-	2791-	2792-	2847-	2848-	2903-	2904-	CQ--A-MG333-3
CQR--A-MG393-3-	2737-	2738-	2793-	2794-	2849-	2850-	2905-	2906-	CQ--A-MG393-3
CQR--A-MG473-3-	2739-	2740-	2795-	2796-	2851-	2852-	2907-	2908-	CQ--A-MG473-3
CQR--A-MG563-3-	2741-	2742-	2797-	2798-	2853-	2854-	2909-	2910-	CQ--A-MG563-3
CQR--A-MG683-3-	2743-	2744-	2799-	2800-	2855-	2856-	2911-	2912-	CQ--A-MG683-3
CQR--A-MG823-3-	2745-	2746-	2801-	2802-	2857-	2858-	2913-	2914-	CQ--A-MG823-3
CQR--A-MG104-3-	2747-	2748-	2803-	2804-	2859-	2860-	2915-	2916-	CQ--A-MG104-3
CQR--A-MG124-3-	2749-	2750-	2805-	2806-	2861-	2862-	2917-	2918-	CQ--A-MG124-3
CQR--A-MG154-3-	2751-	2752-	2807-	2808-	2863-	2864-	2919-	2920-	CQ--A-MG154-3
CQR--A-MG184-3-	2753-	2754-	2809-	2810-	2865-	2866-	2921-	2922-	CQ--A-MG184-3
CQR--A-MG224-3-	2755-	2756-	2811-	2812-	2867-	2868-	2923-	2924-	CQ--A-MG224-3

1/ In the MIL-PRF-19978/11 characteristic K superseding part number, there is no replacement for the .010 to .15 in the 1000 volt dc range. In the MIL-PRF-19978/15 characteristic M superseding part number, there is no replacement for the .0010 to .068 in the 1000 volt dc range. The complete part number will include the applicable symbols to represent style, circuit, tolerance, and product level designator.

2/ Complete dash number will include, as the last digit, the applicable symbol to indicate the capacitance tolerance in percent.

3/ Complete part number will include the applicable symbols to represent style, circuit, and tolerance.

TABLE V-A. Substitutability data.

Superseded dash number of MIL-PRF-19978/1A 3/								Superseding part number 1/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
Style CQ08		Style CQ09		Style CQ12		Style CQ13		
Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
3101-	3102-	3131-	3132-	3161	3162-	3191-	3192-	CQ--A-PC222-3
3103-	3104-	3133-	3134-	3163-	3164-	3193-	3194-	CQ--A-PC332-3
3105-	3106-	3135-	3136-	3165-	3166-	3195-	3196-	CQ--A-PC472-3
3107-	3108-	3137-	3138-	3167-	3168-	3197-	3198-	CQ--A-PC682-3
3109-	3110-	3139-	3140-	3169-	3170-	3199-	3200-	CQ--A-PC103-3
3111-	3112-	3141-	3142-	3171-	3172-	3201-	3202-	CQ--A-PC153-3
3113-	3114-	3143-	3144-	3173-	3174-	3203-	3204-	CQ--A-PC223-3
3115-	3116-	3145-	3146-	3175-	3176-	3205-	3206-	CQ--A-PC333-3
3117-	3118-	3147-	3148-	3177-	3178-	3207-	3208-	CQ--A-PC473-3
3119-	3120-	3149-	3150-	3179-	3180-	3209-	3210-	CQ--A-PC683-3
3121-	3122-	3151-	3152-	3181-	3182-	3211-	3212-	CQ--A-PC104-3
3123-	3124-	3153-	3154-	3183-	3184-	3213-	3214-	CQ--A-PC154-3
3125-	3126-	3155-	3156-	3185-	3186-	3215-	3216-	CQ--A-PC224-3
3127-	3128-	3157-	3158-	3187-	3188-	3217-	3218-	CQ--A-PC334-3
3129-	3130-	3159-	3160-	3189-	3190-	3219-	3220-	CQ--A-PC474-3
3301-	3302-	3329-	3330-	3357-	3358-	3385-	3386-	CQ--A-PE102-3
3303-	3304-	3331-	3332-	3359-	3360-	3387-	3388-	CQ--A-PE152-3
3305-	3306-	3333-	3334-	3361-	3362-	3389-	3390-	CQ--A-PE222-3
3307-	3308-	3335-	3336-	3363-	3364-	3391-	3392-	CQ--A-PE332-3
3309-	3310-	3337-	3338-	3365-	3366-	3393-	3394-	CQ--A-PE472-3

See footnotes at end of table.

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TABLE V-A. Substitutability data - Continued.

Superseded dash number of MIL-PRF-19978/1A 3/								Superseding part number 1/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
Style CQ08		Style CQ09		Style CQ12		Style CQ13		
Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
3311-	3312-	3339-	3340-	3367-	3368-	3395-	3396-	CQ--A-PE682-3
3313-	3314-	3341-	3342-	3369-	3370-	3397-	3398-	CQ--A-PE103-3
3315-	3316-	3343-	3344-	3371-	3372-	3399-	3400-	CQ--A-PE153-3
3317-	3318-	3345-	3346-	3373-	3374-	3401-	3402-	CQ--A-PE223-3
3319-	3320-	3347-	3348-	3375-	3376-	3403-	3404-	CQ--A-PE333-3
3321-	3322-	3349-	3350-	3377-	3378-	3405-	3406-	CQ--A-PE473-3
3323-	3324-	3351-	3352-	3379-	3380-	3407-	3408-	CQ--A-PE683-3
3325-	3326-	3353-	3354-	3381-	3382-	3409-	3410-	CQ--A-PE104-3
3327-	3328-	3355-	3356-	3383-	3384-	3411-	3412-	CQ--A-PE154-3
3501-	3502-	3531-	3532-	3561-	3562-	3591-	3592-	CQ--A-PF102-3
3503-	3504-	3533-	3534-	3563-	3564-	3593-	3594-	CQ--A-PF152-3
3505-	3506-	3535-	3536-	3565-	3566-	3595-	3596-	CQ--A-PF222-3
3507-	3508-	3537-	3538-	3567-	3568-	3597-	3598-	CQ--A-PF332-3
3509-	3510-	3539-	3540-	3569-	3570-	3599-	3600-	CQ--A-PF472-3
3511-	3512-	3541-	3542-	3571-	3572-	3601-	3602-	CQ--A-PF682-3
3513-	3514-	3543-	3544-	3573-	3574-	3603-	3604-	CQ--A-PF103-3
3515-	3516-	3545-	3546-	3575-	3576-	3605-	3606-	CQ--A-PF153-3
3517-	3518-	3547-	3548-	3577-	3578-	3607-	3608-	CQ--A-PF223-3
3519-	3520-	3549-	3550-	3579-	3580-	3609-	3610-	CQ--A-PF333-3
3521-	3522-	3551-	3552-	3581-	3582-	3611-	3612-	CQ--A-PF473-3
3523-	3524-	3553-	3554-	3583-	3584-	3613-	3614-	CQ--A-PF683-3
3525-	3526-	3555-	3556-	3585-	3586-	3615-	3616-	CQ--A-PF104-3
3527-	3528-	3557-	3558-	3587-	3588-	3617-	3618-	CQ--A-PF154-3
3529-	3530-	3559-	3560-	3589-	3590-	3619-	3620-	CQ--A-PF224-3
3701-	3702-	3735-	3736-	3769-	3770-	3803-	3804-	CQ--A-TC152-3
3703-	3704-	3737-	3738-	3771-	3772-	3805-	3806-	CQ--A-TC392-3
3705-	3706-	3739-	3740-	3773-	3774-	3807-	3808-	CQ--A-TC472-3
3707-	3708-	3741-	3742-	3775-	3776-	3809-	3810-	CQ--A-TC822-3
3709-	3710-	3743-	3744-	3777-	3778-	3811-	3812-	CQ--A-TC103-3
3711-	3712-	3745-	3746-	3779-	3780-	3813-	3814-	CQ--A-TC123-3
3713-	3714-	3747-	3748-	3781-	3782-	3815-	3816-	CQ--A-TC153-3
3715-	3716-	3749-	3750-	3783-	3784-	3817-	3818-	CQ--A-TC183-3
3717-	3718-	3751-	3752-	3785-	3786-	3819-	3820-	CQ--A-TC223-3
3719-	3720-	3753-	3754-	3787-	3788-	3821-	3822-	CQ--A-TC273-3
3721-	3722-	3755-	3756-	3789-	3790-	3823-	3824-	CQ--A-TC333-3
3723-	3724-	3757-	3758-	3791-	3792-	3825-	3826-	CQ--A-TC393-3
3725-	3726-	3759-	3760-	3793-	3794-	3827-	3828-	CQ--A-TC473-3
3727-	3728-	3761-	3762-	3795-	3796-	3829-	3830-	CQ--A-TC563-3
3729-	3730-	3763-	3764-	3797-	3798-	3831-	3832-	CQ--A-TC683-3
3731-	3732-	3765-	3766-	3799-	3800-	3833-	3834-	CQ--A-TC823-3
3733-	3734-	3767-	3768-	3801-	3802-	3835-	3836-	CQ--A-TC104-3
3901-	3902-	3915-	3916-	3929-	3930-	3943-	3944-	CQ--A-TE104-3
3903-	3904-	3917-	3918-	3931-	3932-	3945-	3946-	CQ--A-TE124-3
3905-	3906-	3919-	3920-	3933-	3934-	3947-	3948-	CQ--A-TE154-3
3907-	3908-	3921-	3922-	3935-	3936-	3949-	3950-	CQ--A-TE184-3
3909-	3910-	3923-	3924-	3937-	3938-	3951-	3952-	CQ--A-TE224-3
3911-	3912-	3925-	3926-	3939-	3940-	3953-	3954-	CQ--A-TE274-3
3913-	3914-	3927-	3928-	3941-	3942-	3955-	3956-	CQ--A-TE334-3

See footnotes at end of table.

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TABLE V-A. Substitutability data - Continued.

Superseded dash number of MIL-PRF-19978/1A 3/								Superseding part number 1/ MIL-C-19978/1 MIL-C-19978/1B thru MIL-PRF-19978/1F
Style CQ08		Style CQ09		Style CQ12		Style CQ13		
Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
4001-	4002-	4055-	4056-	4109-	4110-	4163-	4164-	CQ--A-TF102-3
4003-	4004-	4057-	4058-	4111-	4112-	4165-	4166-	CQ--A-TF122-3
4005-	4006-	4059-	4060-	4113-	4114-	4167-	4168-	CQ--A-TF152-3
4007-	4008-	4061-	4062-	4115-	4116-	4169-	4170-	CQ--A-TF182-3
4009-	4010-	4063-	4064	4117-	4118-	4171-	4172-	CQ--A-TF222-3
4011-	4012-	4065-	4066-	4119-	4120-	4173-	4174-	CQ--A-TF272-3
4013-	4014-	4067-	4068-	4121-	4122-	4175-	4176-	CQ--A-TF332-3
4015-	4016-	4069-	4070-	4123-	4124-	4177-	4178-	CQ--A-TF392-3
4017-	4018-	4071-	4072-	4125-	4126-	4179-	4180-	CQ--A-TF472-3
4019-	4020-	4073-	4074-	4127-	4128-	4181-	4182-	CQ--A-TF562-3
4021-	4022-	4075-	4076-	4129-	4130-	4183-	4184-	CQ--A-TF682-3
4023-	4024-	4077-	4078-	4131-	4132-	4185-	4186-	CQ--A-TF822-3
4025-	4026-	4079-	4080-	4133-	4134-	4187-	4188-	CQ--A-TF103-3
4027-	4028-	4081-	4082-	4135-	4136-	4189-	4190-	CQ--A-TF123-3
4029-	4030-	4083-	4084-	4137-	4138-	4191-	4192-	CQ--A-TF153-3
Superseded dash number of MIL-PRF-19978/1A 2/ 3/								Superseding part number MIL-C-19978/1 and /1B thru MIL-PRF-19978/1F Style CQ08, CQ09, CQ12, CQ13 1/ 2/
Style CQ04		Style CQ05		Style CQ10		Style CQ11		
Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	Circuit 1	Circuit 3	
4031-	4032-	4085-	4086-	4139-	4140-	4193-	4194-	CQ--A-TF183-3
4033-	4034-	4087-	4088-	4141-	4142-	4195-	4196-	CQ--A-TF223-3
4035-	4036-	4089-	4090-	4143-	4144-	4197-	4198-	CQ--A-TF273-3
4037-	4038-	4091-	4092-	4145-	4146-	4199-	4200-	CQ--A-TF333-3
4039-	4040-	4093-	4094-	4147-	4148-	4201-	4202-	CQ--A-TF393-3
4041-	4042-	4095-	4096-	4149-	4150-	4203-	4204-	CQ--A-TF473-3
4043-	4044-	4097-	4098-	4151-	4152-	4205-	4206-	CQ--A-TF563-3
4045-	4046-	4099-	4100-	4153-	4154-	4207-	4208-	CQ--A-TF683-3
4047-	4048-	4101-	4102-	4155-	4156-	4209-	4210-	CQ--A-TF823-3
4049-	4050-	4103-	4104-	4157-	4158-	4211-	4212-	CQ--A-TF104-3
4051-	4052-	4105-	4106-	4159-	4160-	4213-	4214-	CQ--A-TF124-3
4053-	4054-	4107-	4108-	4161-	4162-	4215-	4216-	CQ--A-TF154-3

1/ Superseding part number will include the applicable symbol to represent style, circuit and tolerance.
2/ Superseded styles CQ04, CQ05, CQ10 and CQ11 are replaced by styles CQ08, CQ09, CQ12 and CQ13 respectively.
3/ Complete dash number will include, as the last digit, the applicable symbol to indicate the capacitance tolerance in percent.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents: In addition to MIL-PRF-19978, this document also references MIL-STD-202 and MIL-PRF-19978/9, /10, /11, /13, /14, and /15.

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2009-045)

Review activities:
Navy - MC
Air Force - 19, 99

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